CIT 4402: Internet of Things

Contact Hours: 45 hours

Prerequisites: CIT 4306: Mobile and Wireless Networks, CIT4303: IT Security and Risk Management, CIT4201: Computer Networks, CIT 4202: Network Design and Implementation

Purpose: The course is designed to provide students with a solid technical introduction to the Internet of Things (IoT).

Expected Learning Outcomes

At the end of this course the student should be able to:

i. Understand where the IoT concept fits within the broader ICT industry and possible future trends.
ii. Differentiate between the levels of the IoT stack and be familiar with the key technologies and protocols employed at each layer of the stack.
iii. Design a simple IoT system comprising sensors, edge devices, wireless network connections and data analytics capabilities.
iv. Use the knowledge and skills acquired during the course to build and test a complete, working IoT system involving prototyping, programming and data analysis.

Course Content

Introduction to the Internet of Things: What is the IoT and why is it important, Elements of an IoT ecosystem, Technology drivers, Business drivers, Typical IoT applications and Trends and implications. Sensors and sensor nodes: Sensing devices and Sensor modules, nodes and systems. Connectivity and networks: Wireless technologies for the IoT, Edge connectivity and protocols, and Wireless sensor networks. Analytics and applications: Signal processing, real-time and local analytics, and Databases, cloud analytics and applications. Industry perspective: Business considerations and Legal challenges.

Mode of Delivery

Lectures, tutorials.

Instructional Materials /Equipment

A computer laboratory, lecture notes, illustration charts, journals, overhead presentation equipment.

Assessment

Type	Weighting (%)
Examination	70%

Continuous Assessment	30%
Total	100%

Core Text Books

i. Dastjerdi, A. V., & Buyya, R. (2016). *Internet of things: principles and paradigms*. Cambridge, MA: Morgan Kaufmann. ISBN: 012805395X.
ii. Chou, T. (2016). *Precision: principles, practices and solutions for the internet of things*. USA: CrowdStory Publishing. ISBN: 1329843568.
iii. Greengard, S. (2015). *The internet of things*. Cambridge, MA: MIT Press. ISBN: 0262527731.

Core Journals

i. *IEEE Internet of Things Journal. ISSN 2327-4662.*
ii. *International Journal of Internet of Things and Cyber-Assurance. ISSN: 2059-7967.*
iii. *IEEE Internet Computing. ISSN: 1089-7801*.

Recommended Textbooks

i. Raj, P., & Raman, A. C. (2017). *The Internet of things enabling technologies, platforms, and use cases*. Boca Raton, FL: CRC Press, Taylor & Francis Group. ISBN: 1498761283.
ii. Ramos, R. O. (2017). *Internet of Things Programming with JavaScript*. Birmingham, UK: Packt Publishing. ISBN: 1785888560.
iii. Sun, J. (2015). *Embedded firmware solutions: development best practices for the Internet of Things*. Berkeley, CA: Apress. ISBN: 1484200713.

Recommended Journals

i. *Internet of Things. ISSN: 2199-1073.*
ii. *International Journal of Internet of Things. ISSN: 2059-7967.*
iii. *International Journal of Web and Grid Services. ISSN: 1741-1106.*

Introduction

Internet of Things (IOT) has many names such as Smart objects, the web of objects, the web of things, and Cooperating objects. Even though there are slight differences in the connotations and definitions of those names, they represent the same fundamental type of technology.

One definition of IOT is a purely technical definition — an IOT device is an item equipped with a form of *sensor* or *actuator*, a tiny *microprocessor*, a *communication device*, and *a power* source. The sensor or actuator gives the IOT device the ability to interact with the physical world. The microprocessor enables the IOT device to transform the data captured from the sensors, though at a limited speed and at limited complexity. The communication device enables the IOT device to communicate its sensor readings to the outside world (to receive instructions, send data or possibly route information) and receive input from other IOT devices. The power source provides the electrical energy for the IOT device to do its work.
For IOT devices, size matters. They are significantly smaller than both laptops and cell phones.
For IOT devices to be embedded in everyday objects, their physical size cannot exceed a few cubic centimeters.
Although this technical definition of an IOT device is important it does not help us understand the behavior, interaction, and other implications of IOT devices. Thus we must define IOT devices based on their behavior.

We already know that IOT devices are able to interact with the physical world by performing limited forms of computation as well as communicate with the outside world and with other similar devices. But *what do IOT devices, given their technical abilities, actually do*?

The answer to this question is not as easy as it seems. First, the behavior of an IOT device depends heavily on where and how it is used. An IOT device deployed in a freighter container to monitor its temperature behaves differently than one that monitors parking spaces. Second, and more important, we cannot know at this point how future IOT devices will be used. Even though we can accurately predict future IOT device uses based on how they are used today, we cannot know exactly what the future usage patterns will be. This is an important point, because it tells

designers of IOT device systems that they must future-proof their systems, protocols, and architectures.

Despite not knowing the exact behavior of an IOT device, there are two behavioral properties common to any one of them: *interaction with the physical world* and *communication*.

IOT devices interact with the physical world by obtaining information from the physical world with their sensors and by affecting the physical world with their actuators. IOT devices use their sensors to sense physical properties ranging from simple and easy-to-measure properties such as light, temperature, and air humidity, to more complex properties such as air pollution, the presence of a car, or when an industrial machine is about to break down. IOT devices affect the physical world using different forms of actuators. This may be as simple as switching on a small LED or as complex as switching on the heat in a particular part of a building.

IOT devices communicate. Even though a single IOT device can be very useful, by turning on the light in a doorway when the door opens, for example, the real power of an IOT device comes from their ability to communicate. The IOT device that would previously switch on the door light is now able to communicate that the door was opened to every other nearby IOT device. These IOT devices may turn on other lights in the house, turn up the heat, and so forth. Likewise, IOT devices in an industrial plant that sense the vibration of machinery may communicate their vibration reading both to each other and to the plant's operator. Communication is essential to the behavior of IOT devices, thus we frequently use the term *networks.*

IOT devices are often embedded into objects (to make them smart) for example, thermometers, car engines, light switches, gas meters. Smart Objects enable many sophisticated applications and solutions

Computing and telephony are two disparate strands of development that have led to the development of IOT. Both computing and telephony play a large part in the formulation of IOT, but the two technologies have different cultural and technical histories.

Currently a number of technologies have contributed to the development of IOT such as: embedded systems, ubiquitous and pervasive computing, mobile telephony, telemetry, wireless sensor networks, mobile computing, and computer networking. Some of these areas come from the computing heritage and some from the telephony heritage. Some have sprung out of academic research communities, some from an industrial background. What they have in common, however, is that they either deal with computationally assisted connectivity among physical items, wireless communication, or with interaction between the virtual and the physical world.

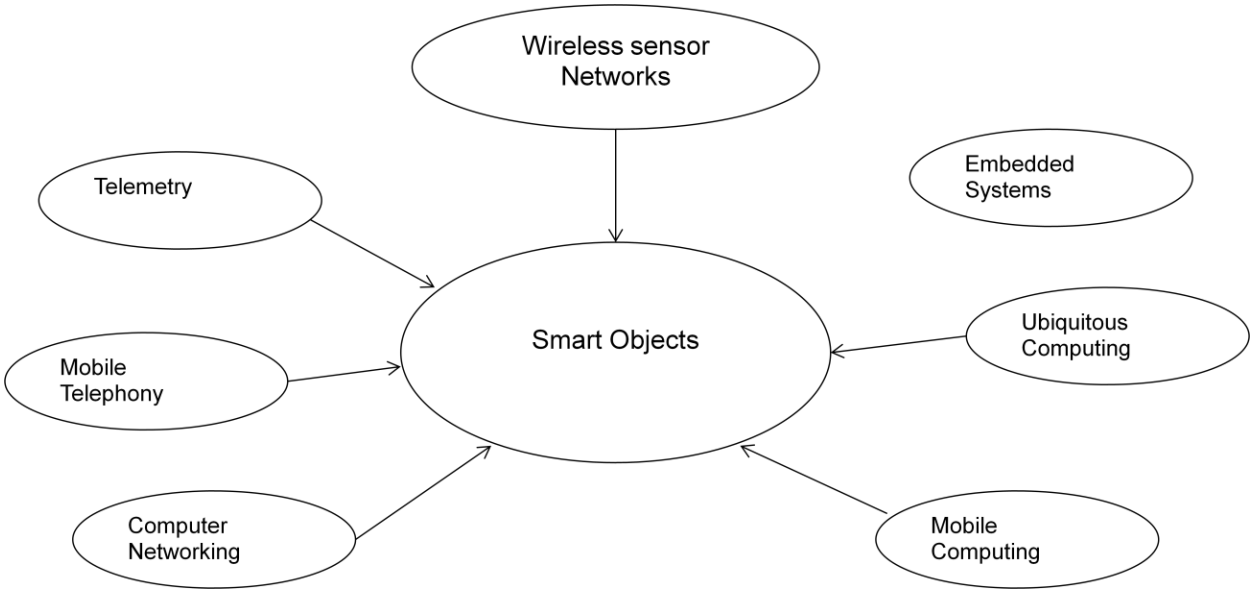


Figure 1: Smart objects are the intersection of embedded systems, ubiquitous computing, mobile telephony, telemetry, wireless sensor networks, mobile computing, and computer networking

IOT ECOSYSTEM

Influential Technologies for the IOT Realization

A number of technologies are necessary in supporting the internet of things. These technologies can be grouped/categorized largely as follows:

Computing paradigms: Service, social, cluster, grid, on-demand, utility, mobile, autonomic, trustworthy, cloud, and fog or edge computing.

Communication: Unified, ambient, and autonomic communication models providing standards-compliant 3G, 4G, and 5G communication capabilities.

Context-aware computing: Ubiquitous sensing, vision, perception, and actuation methods, edge or fog clouds, etc.

Middleware solutions: Integration, intermediation, aggregation, fusion, federation, transformation, arbitration, enrichment, and composition mechanisms.

Digitization and edge technologies: Tags, stickers, smart dust, motes, LED, specks, beacons, chips, microcontrollers, invisible sensors, implantables, wearables, portables, etc.

Sensing, perception, and vision: These are very mandatory for establishing IoT environments.

Miniaturization: Micro- and nanoscale electronics product design techniques and tools.

Knowledge engineering and enhancement: Data to information and to knowledge transition through data mining, analytics, processing, and so on, event processing engines, knowledge discovery and dissemination, knowledge correlation, corroboration techniques, and so on.

Dashboards, report-generation tools, knowledge visualization platforms, and so on play a very vital role in presenting the extracted knowledge in a preferred format.

Interfacing: Natural and adaptive, intuitive, and informative interfaces.

Real-time insights: through in-memory computing and in-database analytics, appliances for real-time processing of IoT big data.

Compartmentalization through virtualization and containerization: The software engineering technique (divide and conquer) is being replicated on hardware resources to bring in the much needed flexibility, extensibility, and maneuverability and ultimately for hardware programming. The hardware components, hitherto black boxes, are being turned and tuned through the smart application of the premier compartmentalization techniques to be white box modules.

Enablers and Barriers of IOT

For the Internet of Things to have the maximum possible impact on economies and society, large-scale adoption is necessary. Adoption depends on having certain technologies, organizational capabilities, and policies in place. It will also require collaboration among supplier companies and among suppliers, users, and policy makers to set the standards and protocols for the interoperability of IoT systems that maximize value. In the table below, we have the technical and non-technical enablers that can maximize IoT impact

Five types of enablers are needed for maximum IoT impact

Software and hardware technology	▪ Low-power, inexpensive sensors and computers ▪ Ubiquitous connectivity/low-cost mesh connectivity ▪ Additional capacity and bandwidth in the cloud ▪ Confidence in security across entire IoT ecosystem
Interoperability	▪ Standardization in the technology stack and ability to integrate across technology vendors ▪ Standard protocols for sharing between IoT systems ▪ Standard access to external data sources
Intellectual property, security, privacy, and confidentiality	▪ Establishing trust with consumers for sharing data ▪ Collaboration across companies and industry verticals ▪ Horizontal data aggregators ▪ Data commerce platforms
Business organization and culture	▪ Industry structure, e.g., organized labor cooperation, third-party servicing ▪ Hardware-focused companies developing core competency in software ▪ Companies committing to up-front investment based on clear business cases
Public policy	▪ Regulation for autonomous control of vehicles and other machinery ▪ Government and payer subsidy of health-care IoT ▪ Agreements on fair practices for data sharing and use

Technical enablers

To achieve the full potential of the Internet of Things by 2025, technical progress is required in three areas:

- improvements in basic infrastructure elements (lower-cost, more capable hardware components and ubiquitous connectivity),
- improvements in software and data analytics, and
- development of technical standards and the technological solutions for interoperability.

Hardware infrastructure

One of the basic requirements of IoT is to have the capacity for millions of devices, machines, and computers to talk each other, sometimes across large distances. For this to happen, two types of base technology are needed to create the infrastructure on which the Internet of Things can flourish: cheap, low-power hardware and ubiquitous connectivity.

- *Low-cost, low-power hardware*

The cost of components and computing power must continue to drop to make IoT applications cost-effective. Today many applications are technically solvable, but the high cost of components such as sensor nodes (with communications and power supplies) makes implementation impractical. However, the declining costs of microelectronics should make critical components more affordable. For example, the cost of semiconductors on a per-transistor basis has fallen by 50 percent in recent years, while the cost of micro-electromechanical systems (MEMS) sensors has decreased by 35 percent. Sensor nodes not only need to be low-cost, but in many remote applications where they cannot be connected to electrical service they will also need to consume little power.

Long-lasting batteries and local power sources (low-cost solar panels) can enable many IoT applications, such as monitoring remote equipment. Low-cost, low-power sensors are also needed in applications such as precision agriculture, where many sensors are necessary for monitoring soil moisture.

The cost of radio-frequency identification (RFID) tags also must drop more to make them practical for tracking low-value inventory in retail, manufacturing, and shipping. EPCglobal, the standards body for the RFID industry, has set a goal to reduce the cost of an RFID tag. Another example application awaiting lower-cost sensors is the smart bandage, which would use disposable humidity sensors to alert patients and caregivers when a wound is not healing properly.

IoT applications will require both low-cost processing power and cost-effective data storage methods. According to IDC, a market research firm, data generated by IoT devices will account for 10 percent of the world's data by 2020, or about 44 zettabytes. Cloud computing and storage pricing demonstrate the ongoing reduction in storage costs—the price of storing a gigabyte of data on a public cloud service fell from $25 cents in 2010 to $.024 cents by late 2014.

Ubiquitous connectivity

Many short-distance connections to IoT sensors will not require cellular data services because the data will travel over low-power local area networks. However, many applications that require more complex

analytic computing of data from diverse sources will need ubiquitous connectivity, which is not yet available, particularly in developing economies. Even in advanced economies, wireless data service can be patchy and unreliable outside urban centers—where many factories, warehouses, and other industrial buildings are located. In the United States, some farmers load sensor data onto USB drives because they cannot count on wireless data networks. In developing economies, coverage in exurban and rural areas is even less advanced, and many countries are still building out high-speed networks. Many developing economies have only wireless data infrastructure (3G/4G/LTE) in urban areas, rather than higher-capacity fiber-optic connections.

Software

The real value of IoT applications comes from analyzing data from multiple sensors and making decisions based on those data. This depends on advances in predictive analytics such as algorithms that can predict a heart attack before it happens based on subtle changes in patient data recorded by home health monitors, or software that can predict when a piece of industrial equipment requires maintenance before it fails.

Today, analytics software has not progressed to the point where it can be easily applied in every case— one reason being that so much of the data that is collected goes unused. The hard work of developing and tuning these algorithms for the peculiarities of specific use cases is largely still undone, and the skills and capabilities to do this work remain in short supply.

There are also gaps in capabilities in user organizations that prevent full implementation of big data analytics in IoT applications. Companies deploying IoT systems need data scientists and managers with skills in data analytics and the ability to customize algorithms. Companies also need access to multiple data sets that reside in silos across the organization.

Interoperability

Interoperability is required to capture nearly 40 percent—and in some cases, 60 percent—of the total potential of IoT. For example, makers of industrial equipment increasingly deliver "connected" products. With built-in sensors and connectivity, these products enable manufacturers to provide anomaly detection and sometimes even predictive maintenance. However, some failure modes can be predicted only when data are combined from multiple systems (predicting over half of the possible failure modes on oil platforms requires data from multiple pieces of equipment, for example). Furthermore, being able to analyze data from multiple pieces of equipment could enable optimization of total processes. But these systems are often not interoperable.

There are many barriers to interoperability, including lack of common software interfaces, standard data formats, and common connectivity protocols. One potential path to overcoming these hurdles is to create common technology standards. Industry associations, technology suppliers, and policy makers can collaborate to create such standards. Translation/aggregation platforms, including common application programming interfaces (APIs), are also needed to manage communication among different applications. They can enable IoT users to extract and manage information from multiple devices and generate the insights (and value) that interoperability provides.

Non-technical enablers

In addition to advances in low-cost hardware, connectivity, and software, achieving the potential of the Internet of Things requires improvements in security and methods for ensuring privacy, protection of intellectual property, and assignment of data ownership. Companies and other organizations that want to maximize the benefits of IoT applications and perhaps build business models around them will need the talent and the proper attitude to manage successful IoT initiatives and make data-driven decisions. Because IoT applications touch on so many areas of regulatory and government responsibility (managing roads and regulating other forms of transportation, for example), policy makers will play a key role in enabling the Internet of Things. Policy makers can also help by addressing concerns about security and privacy and encouraging the development of standards to promote interoperability.

- *Privacy and confidentiality*

The Internet of Things heightens risks to personal privacy and the confidentiality and integrity of organizational data. Some IoT consumer applications, particularly those related to health and wellness, collect sensitive personal data that consumers might not wish to share. Similarly, companies often consider operational data from IoT systems to be highly proprietary.

With IoT applications, consumers may have no idea what kind of information is being acquired about them. To make the in-store offers we describe in the retail environments setting, for example, a retailer would need data about a shopper's previous purchases at that store, the items the shopper viewed at the retailer's website, data about the shopper that the store purchased, and location data about the shopper's path through the store.

Consumers have more than possible embarrassment to worry about from the misuse of private data. Personal data about health and wellness and purchasing behavior can affect employment, access to credit, and insurance rates. For example, underwriting programs that use personal information in addition to credit scores to evaluate applicants for credit could be used in ways that enable companies to discriminate against certain types of consumers.

Many consumers are becoming more wary of sharing personal information with businesses and other institutions. Privacy and security risks to both consumers and organizations will have to be addressed to derive the full benefits from the Internet of Things. Creating compelling value propositions for those whose data are collected and used is critical to adoption. Insurers that use actual automobile use data to evaluate risk, for example, claim that they can reduce premiums by 10 to 15 percent for a majority of consumers. Transparency about data collection and use is also crucial to trust. Finally, protecting the data is essential, as we discuss in the next section.

- *Security*

The Internet of Things heightens existing concerns about cybersecurity and introduces new risks. It multiplies the normal risks associated with any data communication; each device increases the "surface area" available for breaches, and interoperability expands the potential scope of breaches. Every node is a potential entry point, and interconnection can spread the damage. Moreover, the consequences of compromised IoT systems that control the physical world could be catastrophic. A compromised IoT- based home security system or a disrupted medical monitor could pose life-and-death risks. A hacker attack on a smart grid system could potentially turn off power to millions of households and businesses, creating massive economic harm and threats to health and safety. For individuals, IoT security breaches can involve both inappropriate use of personal data and theft.

IoT data security issues involve both inappropriate use and criminal activity (theft)

		Description	**Example**	**Potential ways to address issues**
Legally allowed use of private information by companies in a new way	Loss of control	Data given to corporations cannot be easily deleted	Data cannot be deleted (e.g., data have been resold or are inherently connected to operational data)	**Transparency into collection and use of data, allowing customer to specify permitted uses**
	Imbalance	Companies have detailed information on customers but are themselves not transparent	Detection and use of private information, e.g., pregnancy, by retailers	
	Wrong data forecasts	Analyses have inherent faults, possibly causing discrimination against individual users	Rejection of credit due to duplicate names	
	Discrimination/ exclusion	Companies have the knowledge to identify individual customer groups	Loss of health insurance (e.g., due to detection of high cancer risk)	
Illegal data exploitation by criminals	Multiple scenarios	Illegal exploitation through "back doors" or vulnerabilities	Theft of private information	**Strong end-to-end security**

Mission-critical IoT applications (self-driving cars or military vehicles, for example) should have high levels of security before they are adopted on a large scale. And consumers will not accept applications such as touchless payment systems in retail unless they believe that their payment data are secure. It may be a hard sell: according to the Identity Theft Resource Center, 2014 was a record year for data breaches, with reported incidents rising 27.5 percent from 2013.

- *Intellectual property and data ownership*

A common understanding of who has what rights to data produced by various connected devices will be required to unlock the full potential of IoT. With the exploding growth of IoT data from consumers and enterprises, questions arise about who owns the data. In B2B interactions, data ownership often is not clearly defined or covered in contracts. To capture maximum value and avoid misuse of data, this issue needs to be addressed early in the contract phase of an IoT implementation. In consumer applications, the rights to use data generated in the course of doing business and for what purpose should be clearly spelled out in the agreement that consumers must accept. In cases where more than two parties are involved, industry guidelines or standards should be established that the stakeholders can adhere to.

- *Organization and talent*

Implementing IoT-based systems will require major organizational and behavioral shifts for businesses. The nature and degree of adaptation required will vary by settings and applications. For example, health care remains a high-touch culture—patients expect to interact with a physician. Also, in the United States and elsewhere, physicians are often paid based on the volume of procedures they perform. The Internet of Things could enable a shift toward payment for patient wellness, rather than for treatments. Such a shift may require considerable adjustments by both patients and physicians. More tasks could shift from doctors to physician assistants and technicians. In other industries, IoT raises different organizational challenges. Industries that have unionized workforces (many forms of transportation, for example) may need to negotiate changes in labor practices to enable IoT applications.

Across industries, the most fundamental cultural and organizational shift required for implementing the Internet of Things in corporations is to develop the skills and mindsets to embrace data-driven decision making. Despite the enormous investments that companies have made in data and analytics, many still fail to use data effectively in decision making. According to a study conducted by the MIT Center for Digital Business and the McKinsey Business Technology Office, companies that were in the top third of their industries in making data-driven decisions were, on average, 5 to 6 percent more productive than their peers.

Also, because IoT combines the physical and digital worlds, it challenges conventional notions of organizational responsibilities. Traditionally, the IT organization was separate and distinct from the operating organization that manages the physical environment. In a retail store, for example, the IT function managed the point-of-sale terminals but little else. In an IoT world, IT is embedded throughout store operations, from merchandise tagging to automated checkout. These applications more directly affect the business metrics on which store operations are measured. Adopting the Internet of Things also may require changes in strategy. Equipment makers that use IoT to monitor their machines at customer sites may need to develop new competencies in software and compete on the basis of superior skill in turning IoT data into better service, new products, and new strategies.

- *Public policy*

The digitization of physical systems will require a range of actions from policy makers. Not only will they be called upon to update and strengthen policies to protect the privacy and property rights of businesses and consumers, but they will also need to regulate entirely new forms of activity in the public sphere. Autonomous vehicles are the most obvious example of this challenge. Even as carmakers and technology companies edge closer to road-ready autonomous cars, regulators have not created clear rules that would allow widespread use of these vehicles. Virtually every other form of transportation that can be managed or guided via IoT technology falls under some form of government regulation. Governments even set rules for aircraft and railway equipment maintenance that may need to be modified for usage-based maintenance. Then there are new devices, such as remote control commercial drones, that will require new regulation.

In privacy, security, data ownership, and data sharing, existing regulations will have to be reviewed and updated. Regulators and lawmakers will need the expertise and collaboration of businesses (technology providers and users), citizens, and experts.

Implications of IOT

The Internet of Things enables physical assets to become elements of an information system, creating the ability to capture, compute, communicate, and collaborate in novel ways. The Internet of Things is still in its infancy, but its impact is growing quickly. It is predicted that there will be between 25 billion and 50 billion connected devices by 2025. Governments, policy makers, and businesses have the chance to accelerate the enormous opportunities makers, and the companies that supply IoT technology, beginning with implications for users of IoT associated with the Internet of Things, even as they work to address the risks that accompany the technology. We now look at the implications of IoT for different stakeholders—companies and consumers that use IoT technology, policy technology and policy makers followed by implications for the companies that will supply IoT technologies.

Implications for users and policy makers

As discussed earlier, the introduction of IoT systems into the homes, cities, factories, and other settings has the potential to create value in new ways and to alter how people live and work. IoT applications will affect the full range of stakeholders, from individuals, whose home and work routines will change, to corporations that will have new ways to compete (as well as new competitors). The Internet of Things can enable new business models, including those that monetize the data generated by IoT technology.

- **Consumers**

The Internet of Things offers substantial *benefits* for consumers as well as a new set of *risks*. On the benefit side, IoT applications offer the potential to *drive down the costs* and *enhance the quality of goods and services*, as manufacturers, retailers, and other suppliers employ IoT systems to *improve product design and operations*. Another source of value for consumers will be *greater consumer convenience and time savings*. As they travel, consumers can benefit from IoT-managed roadways, *self-driving cars, real-time public transit information, and planes that take off and land on schedule.* In stores, shoppers could experience a new level of service, savings via *targeted in-store offers, and automated checkout*. At home, they can offload *housework* to smart appliances, *save money on energy* use, and *protect* themselves with IoT-enabled *security systems*. And, most importantly, lives can be extended and improved by IoT *health and wellness applications*.

Given the value associated with interoperability, consumers can choose to purchase systems that are interoperable so that they can maximize benefits. In some cases, this might mean purchasing different systems from one vendor; in others, it might mean purchasing systems from multiple vendors that use common standards.

IoT benefits also come with a new range of risks. The Internet has already raised concerns about consumer privacy, and the mobile Internet (Smartphones) has heightened those concerns by collecting passive tracking data. IoT devices will go beyond that. Wellness monitors can not only report where a person is but also can gather data about what the person is doing. Health IoT systems will collect information about the consumer's weight, eating habits, exercise regimes, drugs used, and diseases. In the workplace, IoT systems can be used to monitor and manage employee

performance, including by "reading" facial expressions to assess a worker's emotional state.

Consumers will need to be diligent about monitoring the data that are being gathered about them. When they sign up for services, they should bear in mind what kind of permissions they are granting for the use of their data, as well as what kind of access they allow to the data they are generating. Consumers can also participate in policy-making dialogues as regulators determine how detailed personal information can be used and how people can exert control over their personal data.

Finally, consumers will have to manage information overload. The physical and psychological effects of constant Internet and Smartphone use are beginning to be acknowledged, as are "context switching costs" when the brain is interrupted by an email alert or an incoming tweet. Consumer IoT systems can be yet another source of overwhelming data and distraction. Consumers will need to think about which IoT systems to adopt and how to integrate the use of those systems into their lives. Many may avoid systems that provide a constant stream of information and instead opt for those that present information only when it is relevant.

- **Employees**

As with other productivity-improving technologies, IoT will affect workers in different ways. Some types of workers are likely to benefit as the Internet of Things creates new needs for specific skills. Workers who have skills in developing and deploying IoT systems will find themselves in greater demand. During the period of rapid growth in IoT deployments, developers and data scientists are likely to benefit the most. Workers with systems integration skills also will be in demand, because of the high degree of customization required for many IoT implementations.

Over time, applications enabled by IoT are likely to enable the automation of an increasing number of tasks currently performed by service workers. For example, IoT could make it possible to economically automate some food preparation. Self-guided cleaning robots could reduce the demand for office and home cleaners, and IoT-based security systems could reduce the number of security guards needed to patrol commercial spaces. Automated checkout sharply reduces the need for cashiers. In general, manual work will come under increasing pressure from IoT and smart machines, but IoT will open up some new employment opportunities, too. Workers who can install and maintain sensor networks, autonomous vehicles, and other IoT systems will be needed.

- **Organizations that use the Internet of things**

The businesses and other organizations that use IoT systems are faced with a multitude of opportunities—and maybe just as many challenges. Depending on the industry, IoT applications can have far-reaching effects on how companies operate and may even force changes in business models. Some businesses (automobile manufacturers, for example) need to master both use of IoT in their operations and in their products. We look at implications for businesses in three ways:

  - using IoT to improve current business models,
  - new business models enabled by IoT and

– Organizational issues related to IoT implementation

i) Using IoT to improve current business models

First, companies need to identify and prioritize opportunities for IoT to enable them to improve performance in their current operations and lines of business. These opportunities are extensions of the opportunities that have been identified with the use of big data and advanced analytics. IoT data will allow companies to personalize services based on consumer behavior, usage, and context. For example, a clothing store that recognizes individual shoppers and knows their purchase histories can suggest outfits that align with their tastes when IoT sensors detect that they have entered the store, saving the consumer from the hassle of browsing through the store's entire inventory. However, even in consumer-related industries, the majority of potential benefits will be in B2B applications, such as manufacturing and supply chain.

It is also important to keep in mind the substantial potential benefits of deploying IoT in businesses in developing markets as well as in advanced economies. Depending on conditions in the specific developing economy, IoT applications can be used to create large amounts of value, often from different sources than those in advanced economies.

Interoperability is key to obtaining much of the value from the Internet of Things. As sophisticated and powerful customers, companies can demand interoperability when they write specifications and procure IoT systems.

As companies identify and prioritize opportunities to use IoT applications, they should inventory all potential sources of data, particularly those they might already own but have not yet fully exploited. IoT is an incredible source of data generated in the course of operations that can be used for other purposes. As noted, in IoT applications, most data, including so-called exhaust data (data that is generated as a "byproduct" of IoT instrumentation), is not yet used or is used only for anomaly detection and/or real-time control. Far more value remains to be captured in analyzing the data for optimization and prediction.

Following are eleven broad categories of applications through which IoT can improve performance and create value for business users across settings and sectors.

The first four categories of applications fall under operations optimization.

- *Inventory management*. Tracking inventory and supplies in retail environments, factories, warehouses, and hospitals
- *Condition-based maintenance*. Deploying sensor data to determine when maintenance is actually needed, reducing breakdowns and costs
- *Human productivity*. Using IoT to teach skills, redesign work, and manage performance
- *Other operations optimization opportunities*. Remotely monitoring and tracking equipment, as well as automatically adjusting machinery based on IoT data, among other uses
- *Sales enablement.* Using IoT usage data to generate new sales
- *Health management*. Improving health and wellness using IoT monitoring data
- *Energy management*. Using IoT sensors and smart meters to better manage energy

- *Environmental management.* Improving stewardship of the environment using IoT technology, such as using sensor data to reduce air pollution
- *Safety and security*. Using IoT sensors to mitigate safety and security risks
- *Product development.* Employing IoT usage data forresearch and development
- *Autonomous vehicles*. Adopting fully or partially self-driving cars, trucks, and public transportation vehicles.

The largest source of potential impact (60 percent) is operations optimization, such as inventory management and condition-based maintenance, which requires optimization and prediction. But each company will have to analyze and prioritize its own opportunities.

ii) New business models for companies that use IoT

In addition to gaining operating efficiencies and the other benefits that companies can derive from using IoT in their existing businesses, companies can create new business models based on IoT e.g.:

- *New pricing models*. IoT creates a continuous, fine-grained stream of real-time data. These data enable both microsegmentation and insights into consumer willingness to pay. Customized, dynamic pricing models based on specific customer preferences, as well as the usage context, can provide benefits for both companies and consumers. For example,
    - auto insurance companies have begun to sell policies with premiums based on sensor data about actual driving behavior, rather than the demographics of drivers.
    - A company selling smart lighting fixtures could charge customers based on the amount of money they save.

- *Service-based business models.* The Internet has enabled the "as a service" business model for IT infrastructure and software. The Internet of Things enables "anything as a service" business models for all kinds of other products, potentially letting many kinds of companies shift from selling products to selling services based on those products. This model can transform large capital expenditures into a pay-by-usage operating expense. Examples of this trend are proliferating. They include selling:
    - "power by the hour" rather than gas turbines,
    - transportation services rather than automobiles,
    - pages printed rather than laser printers.

  IoT technology not only enables the providers of these services to charge by usage, but it also enables these companies to better maintain and upgrade the equipment that is used in these services, removing the maintenance burden from customers and creating the basis of long-term relationships. IoT-based service businesses can also use consolidated customer usage data and data about individual customers for cross- selling, guiding product development, and other purposes. Furthermore, IoT can allow products to become better while in service (the opposite of depreciating in value), a concept Hal Varian has dubbed "product kaizen".

- *Monetization of IoT data*. The exhaust data generated by IoT applications can become a profit center itself. For example, almost any data describing consumer behavior can be of great value to marketers. Data about physical assets such as buildings and vehicles can be used to assess insurance risks. One company's data exhaust could be another company's gold mine, and that value could be monetized by the originator of the data. Of course, privacy, confidentiality, and ownership rights over such data (for instance, consumer purchasing data) are issues that would have to be addressed to create such business models.

iii) Organizational implications of IoT

Companies building strategies around the Internet of Things will need to develop new organizational capabilities and, in many cases, adopt different behaviors. For companies to make the most of IoT technology, their organizations must embrace data-driven decision making and capabilities and create environments in which relevant information is made available to the right decision makers (which could be people or algorithms) in a timely manner. This may require training workers in new skills and investing in new software systems to capture and analyze IoT data. Indeed, all of the implications of big data and analytics apply to IoT data.

Whatever the application, the value of IoT data depends on effective use. Organizations will need to become adept at data-driven decision making, which requires analytical rigor. As companies build up their data analytics capabilities (both technologies and talent) to handle the flood of IoT data, the analytics experts must be connected with decision makers as well as frontline managers; both groups need to learn how to apply insights from large, real-time data streams in a timely manner. And when real-time action needs to be taken on a large scale—such as optimizing machine control across an entire factory or providing real-time marketing offers to shoppers—organizations will have to develop processes that enable algorithms to make decisions automatically, while allowing decision makers to monitor metrics and set policy.

The role of the IT function within organizations will also be transformed by the Internet of Things. The scope of responsibilities of IT departments will no longer be primarily in data centers and managing desktop and mobile devices. IoT applications will extend IT systems into physical assets, facilities, and inventory. Thus, more than ever, the IT organization will have to be closely aligned and even integrated with the operating parts of the business. And the metrics of success for IT will have to measure more than technology, extending to the performance of the business itself.

Synthesizing data from different sources will take more than interoperable systems; it will also require collaboration among leaders and teams in different parts of the organization. Owners of various IT systems (chief financial officers and chief marketing officers, for example) must be receptive to linking up their systems and providing data access across the organization.

Many IoT applications will also require collaboration. For example, in health care, IoT data will need to be shared among doctors, hospitals, insurance companies, device manufacturers, and software providers. In some instances, IoT applications may even depend on competitors working together. In mining, for example, multiple small operators would need to share data in order to optimize their operations in the same way as a large operator. To get insights from data on the characteristics of an ore body in one mine and make effective predictions requires a large comparative data set, which small operators may not be able to get unless they pool

their data.

- **Policy makers**

For IoT applications to achieve their full potential, issues in several areas must be resolved: privacy and confidentiality, intellectual property, security, and interoperability. In each of these areas, government action can be an enabler. The explosion of IoT-based data about what companies and consumers are doing raises important concerns about personal privacy and corporate confidentiality. Who has access to and use of data will become a major issue since many forms of data collection—license plate scanners to catch speeders, for example—do not require consent. Governments can help to establish practices and constraints about data collection, access, and usage, especially for data generated in public spaces. A closely related issue is data rights of various actors in society. What rights do consumers have over data that is generated by or about them? These are also questions in which government has a role to play, through legislation, rule making, policy, and judicial rulings.

Leadership by policy makers can also help address security issues. Governments can invest in research to address security risks, convene and fund multi-stakeholder centers to set standards and share information, model good security practices, and craft thoughtful rules to encourage security management and punish bad actors. Finally, governments and other policy-making organizations, such as industry groups, can encourage and support the development of standards that will enable interoperability of IoT devices and systems.

Implications for technology suppliers

The Internet of Things represents a major opportunity for incumbent technology suppliers as well as for emerging players. The market for IoT components and systems grew 160 percent in 2013 and 2014, and growth is projected to remain above 30 percent a year through 2025. As in other technology markets, the IoT market will have a variety of players and strategies. Some suppliers will compete by offering distinctive technology; others will offer distinctive data. There also will be companies that establish technology platforms and those that specialize in offering comprehensive, end-to-end solutions. The opportunities to assume these roles vary by type of player. We also look at the different sources of competitive advantage for technology suppliers, how we expect value to be distributed among

different types of technology suppliers, and how technology suppliers can play across multiple vertical markets—or find it difficult to do so.

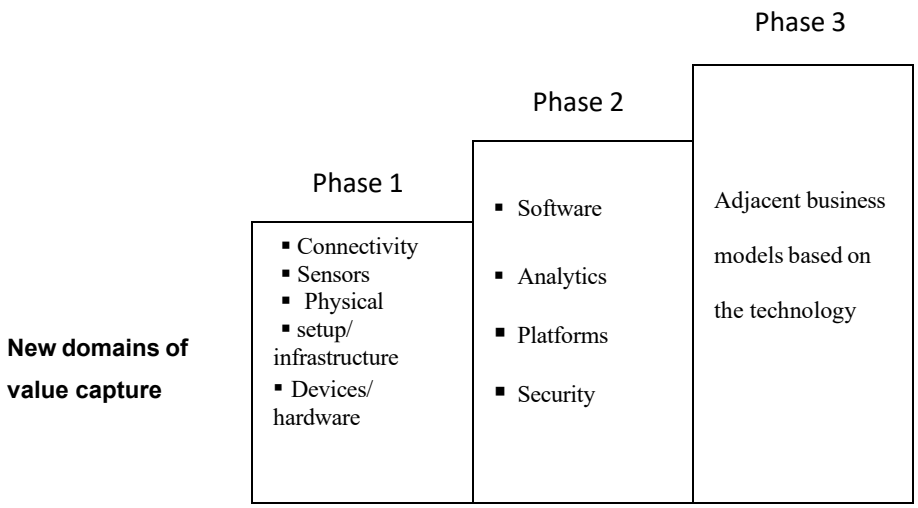
- **The shifting IoT industry value chain**

We expect that, as in other Internet applications, users (businesses and consumers) will capture most of the value created by IoT use—potentially as much as 90 percent in 2025. But given the huge amount of potential value that IoT can generate, this still leaves a substantial market for the suppliers of IoT hardware, software, and services. As the IoT industry evolves over the next ten years, we expect the division of value among players to shift, with an increasing share going to suppliers of software and analytics. Suppliers of foundational technologies such as hardware and IoT device clouds as well as installers of IoT systems are likely to capture less value in 2025 than they do today. Makers of packaged software and applications developers are likely to gain. In 2025, we would expect software and service providers to capture up to 85 percent of the IoT revenue generated by suppliers.

- **Phases in evolution of IoT technology**

The shift in value among IoT suppliers could follow a course similar to the pattern of previous technology waves. In the personal computer industry and in the history of the Internet itself, there were three overlapping periods in which value shifted toward three types of companies. In the first phase, suppliers of infrastructure and hardware dominate. During the Internet era, companies such as Cisco and 3Com provided the basic building blocks of infrastructure. In the second phase, some core services, built on the infrastructure, grow quickly to global scale. In the case of the Internet, companies such as Google and Yahoo provided services such as search. In the third phase, adjacent business models appear. On the Internet, this has included companies such as Amazon, which pioneered Internet retailing, and Airbnb, in peer-to-peer lodging services. Today, the Internet of Things is somewhere between the first and second phases. Some companies are focusing on the basic building blocks, such as connectivity, sensors, and other IoT devices, while others are beginning to specialize in software, analytics, and security that could begin to scale

IoT supplier industry could evolve in three phases



Bases of competition for IoT technology suppliers

For companies to establish a market presence and maintain share over the long term, they will need to understand how their strategies and strengths fit within the three phases of the industry evolution. Across these three phases are at least four distinct bases of competition that can lead to disproportionately large value capture opportunities for individual companies: distinctive technology, distinctive data, technology platforms, and the ability to provide complete solutions. *f* .

- *Distinctive technology*. For broad adoption of IoT technology, several basic technologies are needed, such as low-cost, low-power sensors and low-cost connectivity. Companies that have proprietary intellectual property in these areas can build strong, sustainable positions. In the same way that ARM was successful in the

smartphone era because of its distinctive low-power chipsets, players that develop distinctive IoT-relevant technology will be at an advantage.

- *Distinctive data*. There is significant promise for the owners of valuable proprietary IoT data. For instance, a manufacturer of industrial equipment can collect hundreds of terabytes of streaming data daily from sensors used by its customers, enabling the manufacturer to identify anomalies before they cause equipment shutdowns and power outages, and also enabling the manufacturer to provide predictive maintenance that others with less access to data would be challenged to provide.
- *Platform providers*. In other IT markets, companies have created broad-based software-enabled platforms upon which third parties build applications. Platforms are also a means to provide interoperability among applications. A successful platform exhibits network effects, as each new customer and application adds value for all of the others. Successful platforms also tend to tie the customer to the platform provider, because adopting the platform makes it easy for the customer to take advantage of the entire ecosystem of applications built on the platform. However, large-scale platforms tend to be developed later in the evolution of an industry. It can be quite challenging to scale up platforms before a critical mass of successful use cases and solutions (see next item) has developed.
- *The ability to provide end-to-end solutions*. Because of the degree of customization required to create effective IoT systems in specific industries and companies, companies that can supply complete solutions" — hardware, software, installation, and service — can establish deep relationships with customers that would be difficult for competitors to interrupt. End-to-end suppliers could be makers of equipment that uses IoT technology, software companies, or systems integrators.

The ability to provide complete solutions is one way that IoT suppliers could maintain or expand their position, even as the industry matures and their original advantages become less powerful. Indeed, the shift of value to software does not mean that companies that would traditionally be categorized as hardware suppliers will be at a permanent disadvantage— if they begin to craft solutions that include other parts of the value chain. Increasingly, equipment suppliers are incorporating IoT capabilities in their products and acting as software developers and consultants fortheir customers, thereby providing end-to-end solutions. Similarly, software players are expanding their offerings to include hardware.
In the near term, there will be a premium for providing complex systems integration services, because of the custom nature of many IoT deployments. Systems integration skills are in high demand by IoT users and will likely remain so. Despite some efforts to promote interoperability, it is an extremely complex undertaking to bring together the devices and software that are needed in IoT systems. Moreover, customers need systems that are designed for their vertical markets (clothing store chains, for example) and customized for their specific needs. Today, creating industry-specific or custom IoT systems is labor-intensive and costly. By offering an integrated solution, suppliers can satisfy many customers' desire not to have to piece together solutions themselves (though some sophisticated buyers will). Over time, as the industry matures, we expect more packaged vertical solutions and large-scale horizontal platforms to take hold.

- **Where and how to compete: Horizontal and vertical expertise**

As companies develop IoT offerings, they will need to consider where to compete (what industries or settings to serve)

and how to compete (which hardware, software, and services to offer). One issue that often arises is the degree to which their offerings should address a specific "vertical" niche vs. providing a broad "horizontal" platform. Creating complete vertical solutions requires integration across the value chain and a deep understanding of the industry and setting in which the solution will be deployed. This necessitates considerable investment, both in technology capability (even if done with partners) and in knowledge of the industry and setting. Meanwhile, providing horizontal platforms requires a critical mass of vertical solutions that satisfy specific customer needs in industries and settings. For other IT technologies, there seems to have been a time-dependent element to the evolution, as described above; a critical mass of vertical complete solutions is developed first, and then scaled platforms develop.

For IoT suppliers to grow as effectively as possible, one of their greatest challenges will be to understand which elements of a successful IoT implementation are applicable across different industries and to what degree each vertical and customer will require customization. For IoT suppliers attempting to refine their strategies, this comes down to understanding what is similar (and therefore translatable) among industries and specific customers and what differs.
As this point, we observe that while there are substantial opportunities for shared technology and capabilities across customer industries, the ability to offer general-purpose IoT software products is limited. Companies seeking to offer products across multiple industries will need to either specialize in very basic technology, such as hardware or connectivity, or develop industry expertise for each vertical they pursue as well as the capacity to customize vertical products for specific submarkets and customers. The opportunity for cross-vertical application of IoT technologies varies by the level of technology

While some elements of IoT software are easily shared—device management, data management, and security, for example—using the same algorithms to interpret data across different deployments is challenging. For example, even though many IoT applications need algorithms to perform common tasks, such as detecting patterns in streams of data, considerable tailoring is necessary to use the same algorithm in multiple contexts.

One widely used type of algorithm is for pattern recognition. This is basic to condition-based maintenance and health monitoring applications, both of which depend on determining the patterns in variables that are predictors of failure. The nuance driving these combinations can require a high degree of subject matter expertise to accelerate the process of creating predictive algorithms. Predicting the failure of an engine might require a combination of pressure, temperature, and engine vibration data. A system that can issue an alert only when readings on a mining truck are out of range might issue hundreds or thousands of alerts in a month, rather than identifying the handful that, when they appear in combination, indicate a need for preventive maintenance.
Even within an industry, a single algorithm that is not tailored to specific settings has limited application. ConocoPhillips spent five years developing an advanced software system to manage its complex of wells in the Ekofist field off the coast of Norway. However, translating that system to other locations even within ConocoPhillips would require substantial customization since some of the algorithms are specific to the geology of the Ekofist region. In some cases, transferring capabilities (the talent and experience of experts in IoT implementation, for example) is easier than transferring technology.

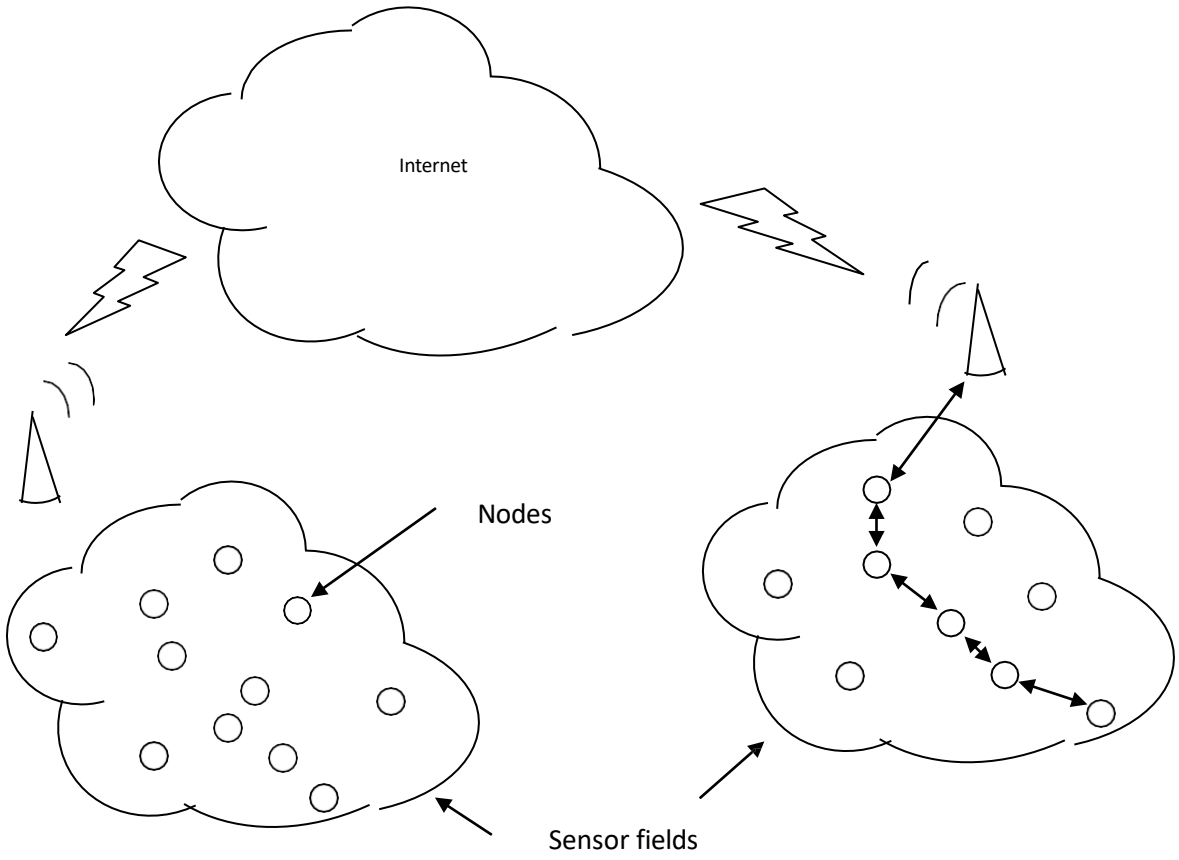
However, there are examples of algorithms that could be used in different IoT implementations, a few of which we describe here:

- *Pattern recognition of stream data*. Most of the routines used to find patterns in streams of data from IoT devices rely on well-known statistical techniques. The patterns that predict whether a truck engine or a gas turbine will break down are relatively similar. However, there are substantial differences that can require months of work by specialized data scientists and subject matter experts to tailor for specific applications. A new generation of algorithms being developed attempts to overcome this limitation using techniques from topology (the mathematical study of shape) and neural networks, especially deep learning.
- *Resource allocation*. Resource allocation problems (where to deploy employees, equipment, and other assets) are similar across settings, providing a way to maximize a specific outcome (plant output, for example) given a number of constraints (such as number of employees and time). The algorithms and math behind the optimization share a lot of similarities across settings, even when problems are large—some involve 20,000 constraints or more. The biggest difference between applications is calculating the impact of each constraint on the goal. Some impacts can be estimated using statistics and big data (how much a reagent increases yield in a pharmaceutical manufacturing plant, for example); others, however, will require knowledge of the business context and subject matter expertise (the additional value of assigning another employee to a construction site, for example).
- *2-D layout optimization*. Placement of goods and equipment in a 2-D space (a shop floor, for example) is a fairly common problem. The algorithms for each type of shop share a lot of similar characteristics. 2-D layout is a more specific version of the broader constrained optimization problem that resource-allocation algorithms attempt to solve.
- *3-D layout optimization*. In 3-D layout optimization, the challenge is to find the best way to pack a space, and the problem is far more complex. Maximizing the storage capacity of a truck or shipping container requires a very different set of algorithms than would be used to optimize placement of cereal boxes on a store shelf.
- *Path routing*. Path routing is used to determine how and where goods, equipment, and people should move most efficiently. This is one of the broadest algorithmic problems for IoT applications and is used in routing trucks, or employees in a warehouse. All share a number of similar elements. Given the large number of possible paths (to get from one part of New York to another that is 15 blocks up and 15 blocks over, there are 155 million possible paths), the common challenge is to select a smaller set of alternatives to make the computation efficient.
- *Computer vision and hearing*. Object recognition is the computer vision problem of distinguishing one object in an environment from another (identifying a table or a chair in a room). The underlying algorithms for computer vision are relatively similar, but a high degree of customization might be required if a system needs to be better at distinguishing a specific type of object. Distinguishing a dog from a cat, for example, can require much more customization than distinguishing a dog from a table. Sound analysis, for recognizing speech and other sounds, has similar requirements. Deep learning has also been successfully applied to these problems.
- *Emotion/mental state analysis*. Emotion analysis algorithms attempt to analyze the mental and emotional state of humans through computer vision and sound analysis. While the underlying algorithms can be built on top of

those used for vision and speech, customization is required to adjust for context. The degree of customization varies. A safety system that needs to determine merely if a person is tired can be much more accurate with some customization than a general algorithm that needs to determine someone's broader emotional state, for example, to help a customer service representative to respond in an optimal way. The more emotional states a system needs to distinguish, the greater the nuance and customization required. Similarly, the more specific a distinction needs to be, the more customized the algorithm will need to be (distinguishing between excited and nervous for a casino is more complex than distinguishing between happy and sad for a retail store).

Wireless Sensor Networks

- A wireless sensor node consists of sensing, computing, communication, actuation, and power components. These components are integrated on a single or multiple boards, and packaged in a few cubic inches.
- Sensor networks are used for communications among small-sized, low-cost, and low-energy- consuming devices (sensors) for which a high data rate is not necessary.
- sensor networks must be able to self organize
- WSN usually consists of tens to thousands of nodes that communicate through wireless channels for information sharing and cooperative processing.



Most sensor networks use radio communication, even if alternative solutions are offered by laser and infrared. Nearly all radio-based platforms use COTS (Commercial Off-The- Shelf) components.

- Sensor nodes are partitioned roughly into four classes based on their physical size:
    - Brick

- Matchbox
- Grain
- Dust.
-

- In ad hoc networks, nodes can directly communicate with each other without an infrastructure. Nodes may act as routers, forwarding messages over multiple hops on behalf of other nodes.

Types of sensors

- Acoustic
- Thermal
- Hydrophones
- etc

Topologies

- star
- mesh
- One way
- Bi-directional

Typical sensor network application

- Environment monitoring
- Chemical and biological sensors
- Fire and earthquake emergencies
- Vehicle tracking for traffic control
- Surveillance of city districts
- Defense-related networks
- Alerts to terrorist threats

Differences of sensor networks in comparison to traditional ad-hoc networks

- Number of nodes that compose a sensor network is usually much bigger than the ad-hoc network one.

- Sensor nodes experience failures.
- Sensor network topology changes more frequently.
- Sensor networks use broadcast communication whereas ad-hoc net- works tend to rely on point-to-point communication.
- Sensor nodes have limited resources (power, memory, computational).

Design of Sensor networks has to take into account several aspects that can affect performance such as:

- The system has to be resistant to node failures (fault tolerance): when a node experiences a failure due to hardware damages or lack of power, the whole system must be able to continue efficiently.
- Management schemas have to face problems caused by the great number of nodes that characterize networks. Nodes must consume very low power, must be able to adapt to different environments, must be autonomous and must have low production costs.

Open problems regarding sensor networks

- In the physical layer most problems are related to energy consumption and frequency modulation issues
- Data the link layer: related to error coding control, power-saving operations, and Medium Access Control schemas.
- Network layer: Routing algorithms
- Transport layer: Design of protocols able to support the particular nature of sensors
- The application layer: the more unexplored layer despite some protocols for sensor management proposed.

Wireless Technologies

The scope of application or usage of an IoT network has a significant impact on the choice of the optimal solution in terms of adequate hardware and software infrastructure. The wide scope of IoT usage, where each of them specifically dictates hardware, software and communications needs, defines the factors that must be captured by the analysis process. The choice of a particular technology depends on the scope of the IoT network because it can significantly affect the design process.

A number of features that have a significant impact on the process of designing and implementing an IoT network include:

- Reach
- Bandwidth
- Coverage
- Power consumption
- Reliability
- Latency
- Cost-effectiveness

Range

The application of IoT in smart city areas on the one hand, as well as smart meter readings on the other, leads to conflicting requirements regarding the reach of a communications network and can extend from a few kilometers to just a few centimeters of distance. The example above explains the importance of range as an important criterion for choosing the right IoT network implementation. Depending on the distance of IoT nodes, communication networks are divided into three basic categories:

- short-range networks
- medium-range networks
- and long-range network

Bandwidth

Depending on the application of the IoT network, nodes may send information periodically, e.g. in smart home systems, fire detection sensor connects to the network and sends information in case smoke/fire is detected. In contrast, if IoT nodes are used in security applications such as security cameras, then there must be constant communication and the transmission of significant amounts of data. Increasing bandwidth affects the reach and power consumption, which entails other factors such as the cost and cost- effectiveness of the entire system, making bandwidth an important criterion for making the right decision when choosing an IoT network implementation.

Reliability

If the designed IoT system is used in medical systems to monitor the patient's vital parameters, one of the most important features is that the correct information is successfully transmitted to the destination as soon as possible. On the other hand, in the case of the application of the IoT system in agriculture, the transmission or omission of transmission of information on e.g. soil humidity at a certain interval, will not cause significant damage. Increasing reliability also has a significant impact on increasing bandwidth and

cost, which puts reliability at the center of an important criterion for choosing an IoT network implementation

Latency

Network latency is defined as the time it takes for a request to be forwarded from the transmitting device to the receiving device, including the time it takes the destination device to process it. In cases of application of IoT nodes in industry, sending control information and signals, e.g. for the management of the power grid, it can be concluded that in this case, it is very important that the information is transmitted to the destination so, if a problem occurs, then the network can be switched off as soon as possible. In contrast, the application of the IoT network in agriculture, e.g. soil temperature measurements, the transmission time can be considered negligible. Reduced latency has the effect of reducing the lifespan of autonomous IoT nodes, and is, therefore, an essential factor in the correct choice of IoT network implementation

Energy efficiency

Energy efficiency, that is, the energy consumption of IoT nodes can certainly be seen as one of the key parameters when designing an IoT network. The concept of IoT itself encompasses autonomous sensor nodes that have an increased lifespan, which is a considerable problem in the development process. By introducing state-of-the-art technological solutions into the process of developing hardware components of IoT, the problem of autonomous power sources has been significantly resolved. However, while the capacity and quality of autonomous power sources are increased, energy efficiency remains a key element in the design process. Some of the reasons that lead to increased consumption are the constant activity of IoT nodes and their sensor units, as well as the large exchange of data within the sensor network. Minimizing the activity time of IoT nodes, eliminating full-duplex or half-duplex communication models, and reducing data transmission time as well as their amount (encryption application and hexadecimal format transfer) have made significant savings in terms of energy efficiency. Also, when designing IoT nodes, it is necessary to pay attention to the consumption of sensor elements that can significantly cut the life of an IoT node. The configuration of nodes in the network, as well as the atmospheric conditions of the environment in which the IoT nodes are located, can also affect energy consumption.

Based on the above, it can be concluded that low power consumption in wireless IoT networks is a feature that is possible and feasible, but to a considerable extent limited

Coverage

Wireless coverage is defined as the extent of the area to which the wireless signals are transmitted, and generally can be divided into five categories that differ in distance and speed of access

- Wireless Body Area Network (WBAN) - used to monitor and collect vital parameters of a person,
- Wireless Personal Area Network (WPAN) - Provide short-range communication for devices.
- Wireless Local Area Network (WLAN) - provide communication in a restricted area.
- Wireless Metropolitan Area Network (WMAN) - networks that provide communications at the city level.
- Wireless Wide Area Network (WWAN) - connect large geographical areas (countries or continents).

The challenge of picking the right communication technology for the implementation of the IoT network depends largely on the application of the designed IoT system. For instance, the application of IoT in designing smart cities, or smart agriculture, coverage differs significantly from the application of IoT for medical purposes. In these cases, coverage can range from a few centimeters, up to several tens of kilometers. It is important to note that greater network coverage does not necessarily mean better data flow or speed of exchange. Coverage is a necessary but not sufficient condition for data transmission and sharing to be possible

Cost-effectiveness
Cost-effectiveness is an aggregate criterion that is closely related to the business model and is proportionally dictated by the budget, needs, and capabilities that the designed IoT system must have. Cost-effectiveness is calculated according to all other criteria, e.g. systems where large amounts of data are transmitted over the network, depending on the topology of the network nodes (if the nodes are at a shorter distance from the system's data concentrator and close to the power source), Wi-Fi could be the best choice. If it is a WAN system, then one network technology that works on the principle of mobile data will be an adequate choice. The selection of the appropriate network should primarily comply with the basic requirements of the system to select the network that will have optimal cost-effectiveness for a particular IoT system.

Technologies

Wireless communication technologies are defined as the ability to transmit data or information wirelessly from one device to another, and as such are an integral part of any IoT system. Depending on the domain of application of the IoT system, information is transmitted from one device to another at different distances that can range from a few centimeters to several hundred kilometers through clearly defined transmission channels.

Due to the heterogeneity of the market, as well as the technologies available today, there is no unique technology for wireless transmission, and most often they can be divided by the length of the electromagnetic waves and the frequencies at which they operate. Figure 1 and Table 1 below depict an IOT general architecture and the comparative statistics respectively.

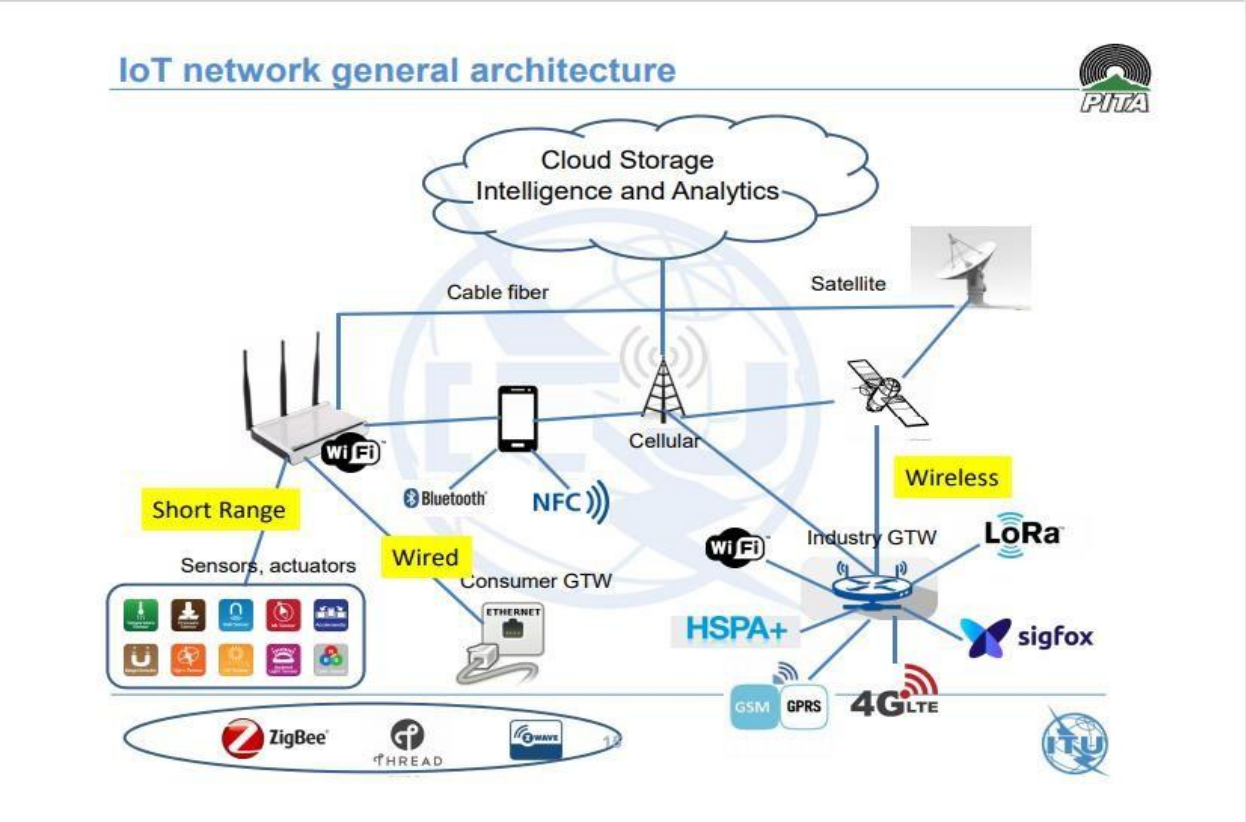


Figure 1

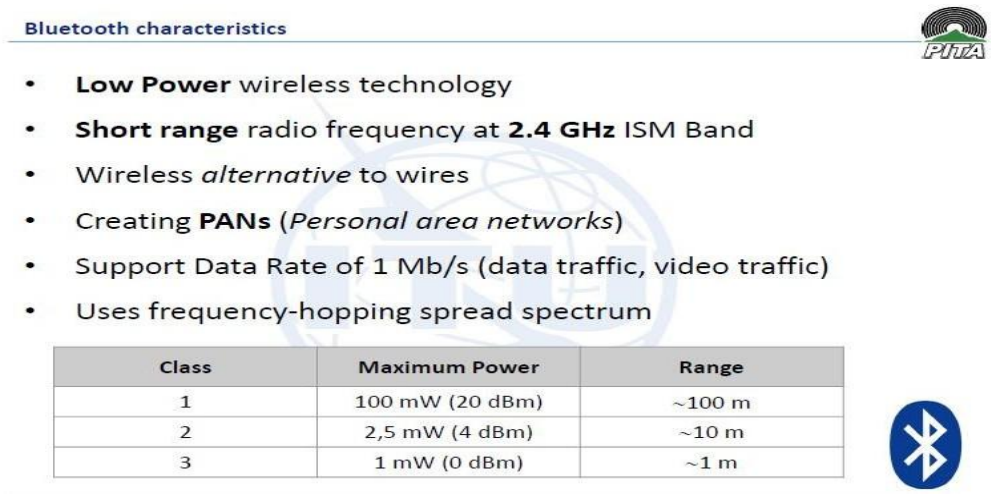
TABLE I. COMPARATION OF WIRELESS COMMUNICATION TECHNOLOGIES

Wireless Communication Technologies	Parameters							
	Range	*Bandwidth*	*Reliability*	*Latency*	*Power use*	*Coverage*	*Cost per module*	*Example of use*
Bluetooth Low Energy	< 10 m	< 2.4 GHz	Medium	< 10 ms	Low	Small	$2 – $5	Blood pressure measurement
Near-field Communication (NFC)	< 4 cm	13.56 MHz	High	100 ~ 250 ms	Very Low	Very small	Reader $5 - $40 Tag < $1	Mobile payments
ZigBee	< 100 m	< 2.4 GHz	High	< 20 ms	Low	Small	$1 - $5	Ground parking sensors
Wi-Fi	< 100 m	2.4 GHz – 5 GHz	High	< 10 ms	Medium	Small	$1 - $10	Remote control robot car
LoRaWAN	5 km urban 20 km rural	500 KHz – 900 MHz	Medium	1 ~ 10 s	Low	Large	$2 – $25	Smart trash bins
Narrow-band IoT (NB-IoT)	> 10 km	450 MHz-3.5 GHz	High	1.5 ~ 10 s	Low	High	$5 – $30	Smart warehouse
SigFox	10 km urban 50 km rural	100 Hz – 900 MHz	Medium	1 ~ 30 s	Low	Large	$4 – $60	Smart wineyards
3g/4g/LTE	> 30 km	200 KHz – 900 MHz	High	100 ms ~ 1 s	Low-High	Large	$4 – $35	Forest fire detection

Some of the common technologies used currently are described below.

A. *Bluetooth*

Bluetooth is a type of wireless connection developed by the Bluetooth Special Interest Group (SIG). The basic purpose is communication between devices at short distances and is most commonly used to connect a computer peripheral. It operates in the 2.4 GHz ISM (industrial, scientific and medical) band, and more recently it is the de-facto standard in the mobile industry. Its characteristics are summarized in figure 2 below.



Class	Maximum Power	Range
1	100 mW (20 dBm)	~100 m
2	2,5 mW (4 dBm)	~10 m
3	1 mW (0 dBm)	~1 m

Figure 2

Bluetooth Low Energy (sometimes referred to as Bluetooth Smart) is a variant that was launched in 2011 as part of the Bluetooth 4.0 specification, and the advantage over earlier versions is that it uses less power, has less setup time, and supports an unlimited star type topology by the number of nodes. BLE is specific for devices with low bandwidth, as well as for devices powered by autonomous power sources. BLE characteristics are summarized in figure 3.

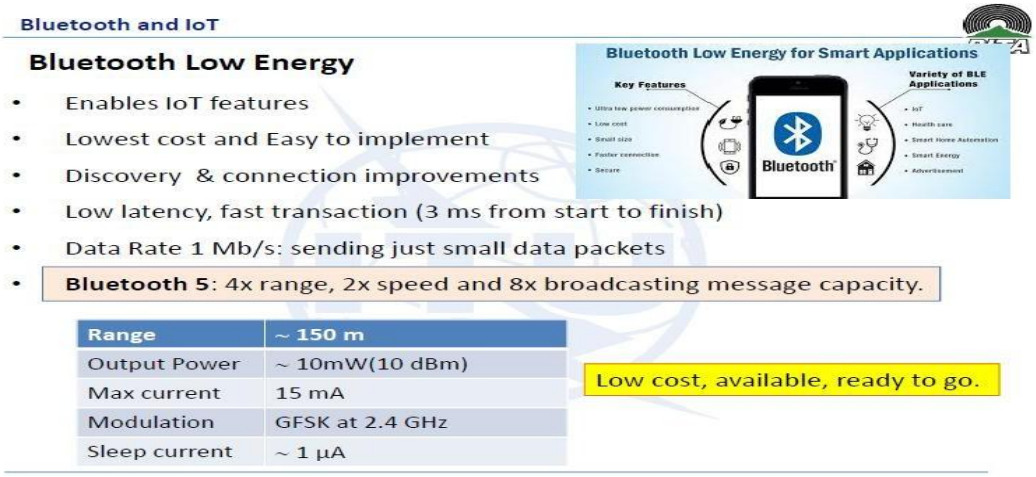


Figure 3.

B. Near-field Communication (NFC)

NFC is a wireless communications technology that was developed by Sony and Philips. It works on the principle of magnetic coupling between the devices and enables two-way data transfer when the two devices are placed side by side (up to a maximum distance of 10 cm). NFC is RFID based and operates at up to 13.56 MHz. Includes a tag that contains a small amount of data, and can be read-only or read-write when the device rewrites the data. The most common application is in mobile devices for contactless payment, that is, storage and access to all types of personal data

C. Wi-Fi

Represents the most commonly used communication standard, assigned as 802.11 by the Institute of Electrical and Electronics Engineers (IEEE). The technology is promoted by the non-profit Wi-Fi Alliance, which certifies products with WiFi interoperability. It works in the 2.4 to 5 GHz range, and an access point and a wireless network adapter are required to set up a core network of this type. Wi-Fi in its reach, data transfer speed, and market penetration is a key technology available in the market today. Figure 4 depicts the characteristics of Wi-Fi variants.

WiFi

- Wireless technology
- Alternative to Wired Technologies
- IEEE 802.11 standard for WLANs

Standard	Frequency bands	Throughput	Range
WiFi a (802.11a)	5 GHz	54 Mbit/s	10 m
WiFi B (802.11b)	2.4 GHz	11 Mbit/s	140 m
WiFi G (802.11g)	2.4 GHz	54 Mbit/s	140 m
WiFi N (802.11n)	2.4 GHz / 5 GHz	450 Mbit/s	250 m
IEEE 802.11ah	900 MHz	8 Mbit/s	100 M

Figure 4

Wi-Fi is used widely with different devices as depicted in figure 5.

Figure 5: Wi-Fi devices

D. Cellular 3G/4G/5G LTE

The cellular network is a large area covered by radio waves that are divided into defined shapes and sizes called cells. Each area has one transceiver (base station), which connects one device to another in the same or different cell. Such networks offer low energy consumption and a large coverage area. The cellular network first appeared in the 1990s under the designation 1G (First generation). Since then, up to the present day, new versions (2G, 3G, 4G) are emerging, as well as inter-versions, which differ in characteristics, primarily by transmission mode (analog-digital) and power consumption. 5G and LTE (Long term Evolution) represent the latest generation of cellular technologies and are popular with mobile phones and IoT devices that require high bandwidth. 5G networks, compared to the previous generation, provide more speed and more channels, which results in increased capacity and allows more devices to connect.

E. LPWAN – Low Power Wide Area Networks

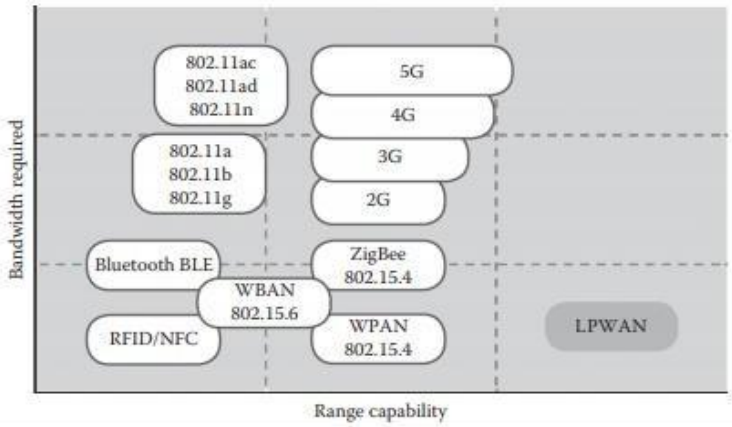
Low power wide area networking technologies (LPWAN) technologies are used extensively for communication in the IoT ecosystem. LPWAN technologies are superior when compared to Bluetooth and BLE for M2M communication because of their cost effectiveness and low-power consumption. LPWAN technology is ideal for connecting devices that send small amounts of data over a long range with battery efficiency. An ideal example of devices that fall under this category are sensors, which are used for transmitting data within smart homes, buildings, parking systems, etc.

The key features of LPWAN that make it suitable for IoT ecosystem are the following:

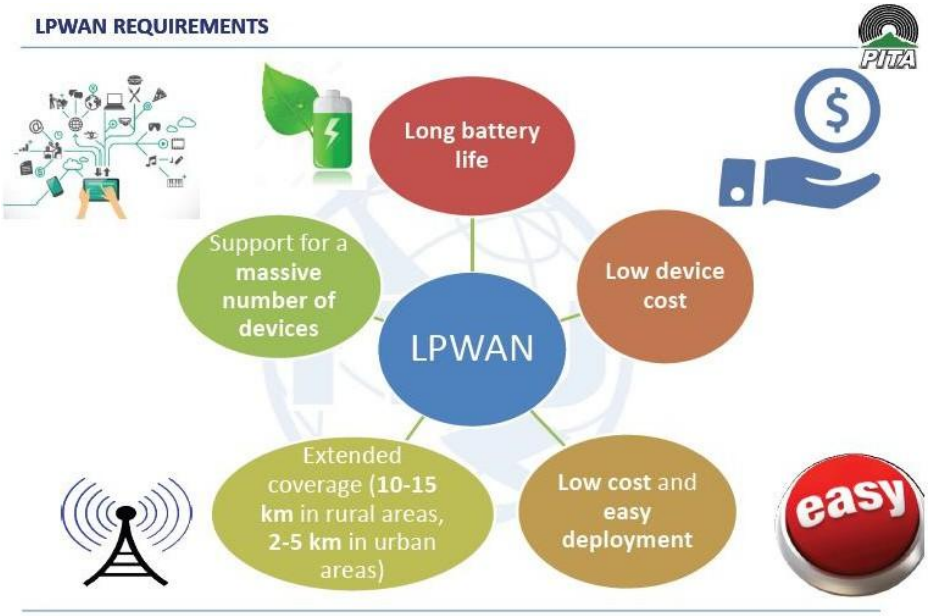
- Long-range communication: Ability to support nodes that are greater than or equal to 10 km distance from the gateway. However, the correct distance is based on the LPWAN technology that is used.
- Low transmission data rate: Less than 5000 bits of data are sent per second. Often only 20–256 bytes per message sent several times a day.
- Low-power consumption: This provides very long battery life for the devices. Many times, the battery life may last up to 10 years.

LPWAN technology is ideally suited for the following two types of applications:

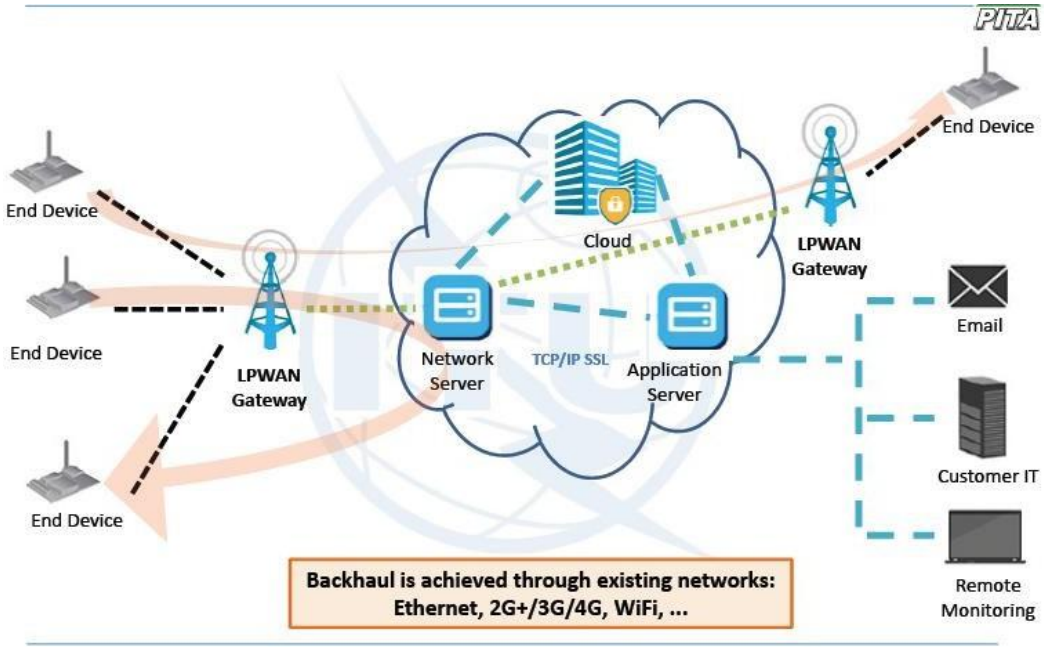
1. Fixed, medium-to-high density connections: This is mainly used in cities and for buildings as an alternative option for cellular communications. Some common examples are smart grids, GPS-based asset tracking systems, and smart lighting systems.
2. Long life, battery powered applications: For applications that need a long range. Some exam ples are water meters, gas detectors, smart agriculture systems, etc.



Bandwidth requirements of LPWAN technologies



LPWAN general Requirements



General architecture of LPWAN

F. SigFox

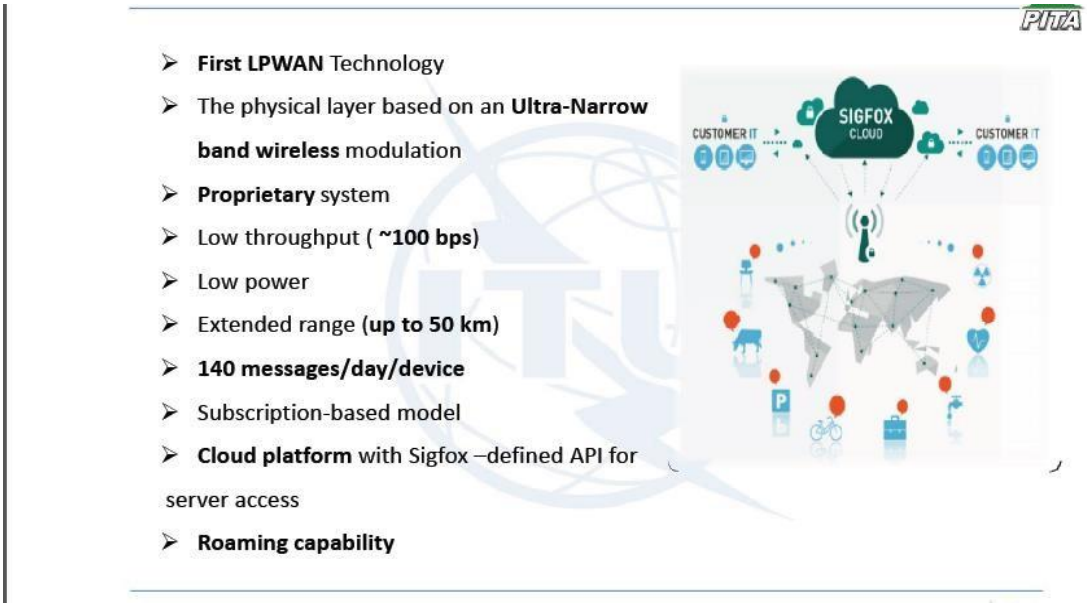
SigFox LP-WAN is an end-to-end system consisting of the presence of a certified modem and ending with a web-based application. Developers must acquire a modem from a certified manufacturer to integrate into their IoT endnode device. Alternatively, there are third-party service providers that make SigFox-compatible access point networks available to handle traffic between the endnodes and SigFox servers. The SigFox servers manage the endnode devices in the network, collect their data traffic, and then make the data and other information available to the users through a web-based API.

To provide security, the SigFox system uses frequency hopping, which mitigates the risks of message interception and channel blocking. Furthermore, SigFox provides anti-replay mechanisms in their servers to avoid replay attacks. The content and format of data sent in the transmission is user-defined so only the user knows how to interpret their device data.

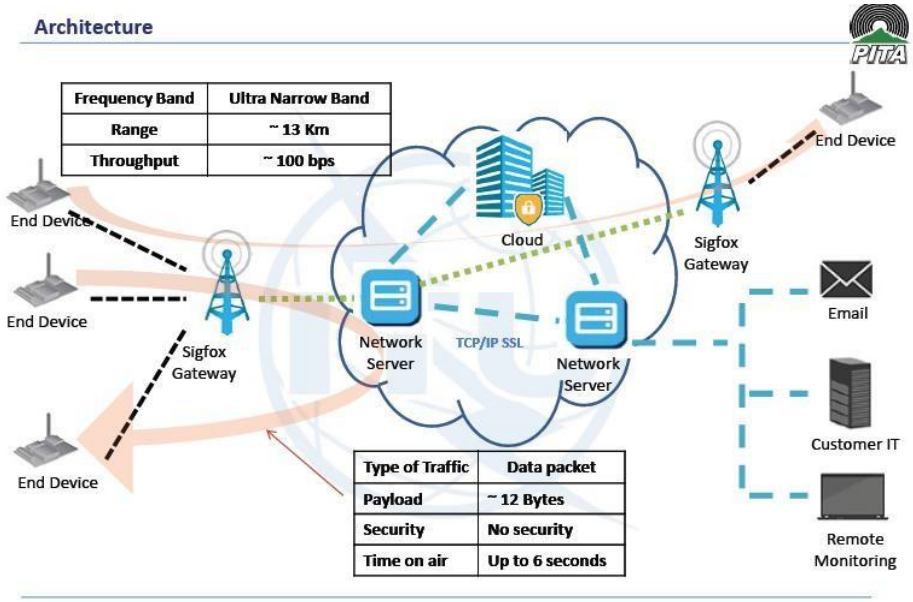
Frequency band: 868MHz/902MHz ISM

Channel width: Ultra narrow band

Range: 30-50km (rural), 3-10km (urban),



Sigfox Overview



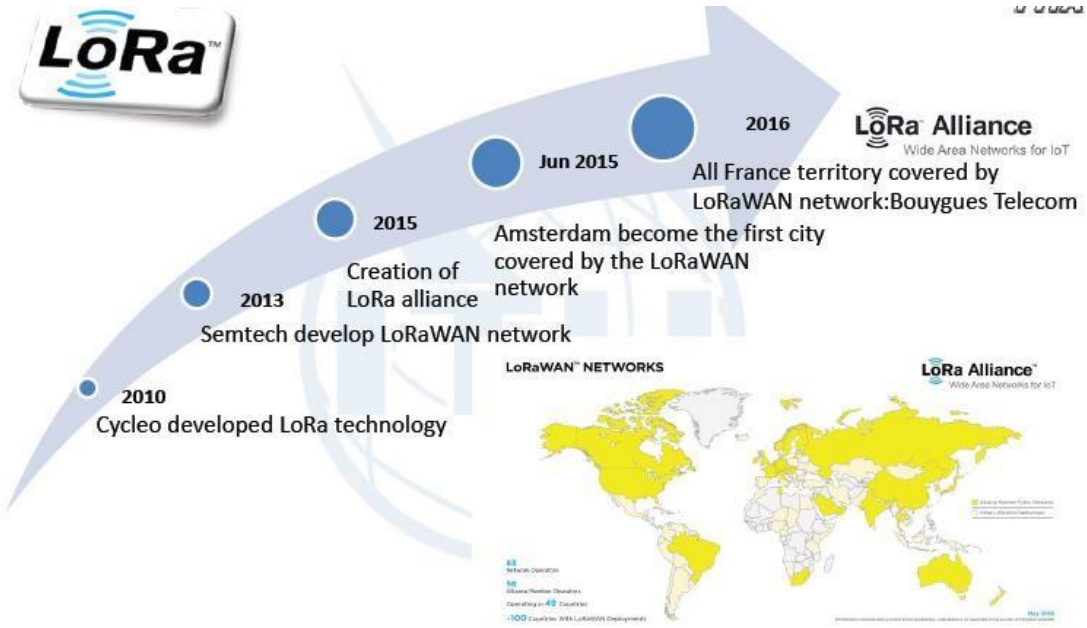
Sigfox Architecture

G. LoRaWAN

The LoRaWAN architecture is a "star of stars" structure of endnode devices connecting through gateways in order to connect to network servers. The way LoRaWAN works is that the wireless hop between the endnodes and gateway uses a Semtech proprietary spread spectrum radio scheme, which they call a *chirp.* The endnodes communicate by chirping the gateway when they have data to transmit. The radio scheme allows the network server to manage the data rate for each connected device, which allows the LoRaWAN connections to decide between payload and range dependent on local radio conditions.

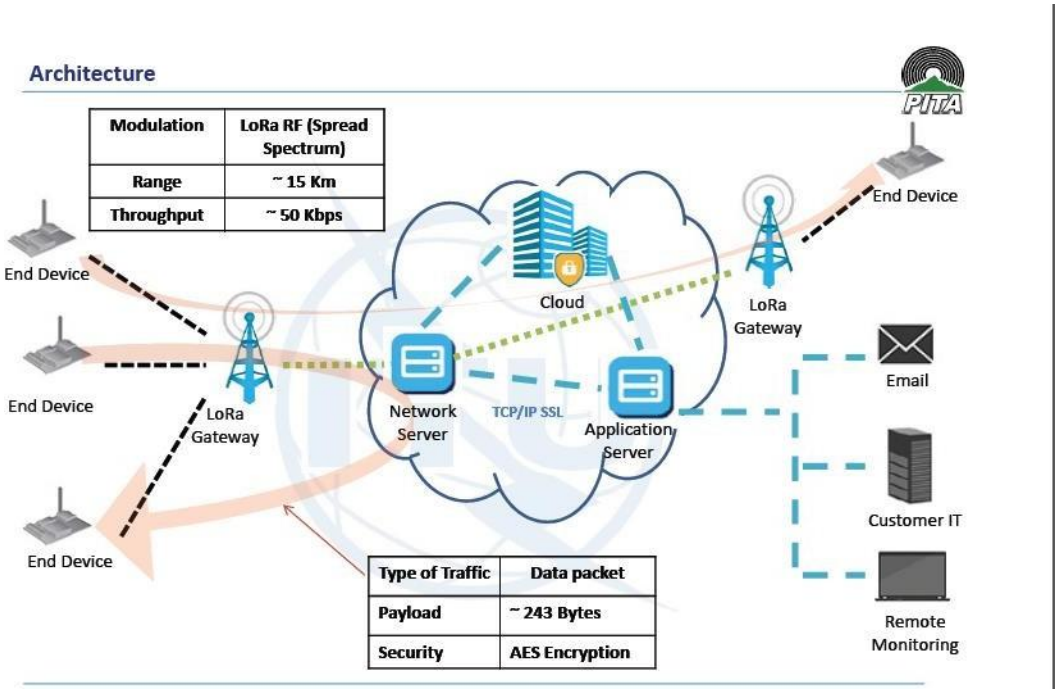
The LoRaWAN network topology has three classes of endnode device. Class A are bidirectional devices, which have a scheduled uplink transmission window and two, short downlink receive windows. Class B devices have additional, scheduled downlink windows and Class C devices have open receive windows. Security for LoRaWAN includes the use of unique networks and device-encryption keys.

Frequency band:	433/868/780/915MHz ISM,
Channel width	EU: 8x125kHz, US 64x125kHz/8x125kHz
Modulation:	Chirp Spread Spectrum
Range:	2-5k (urban), 15k (rural)

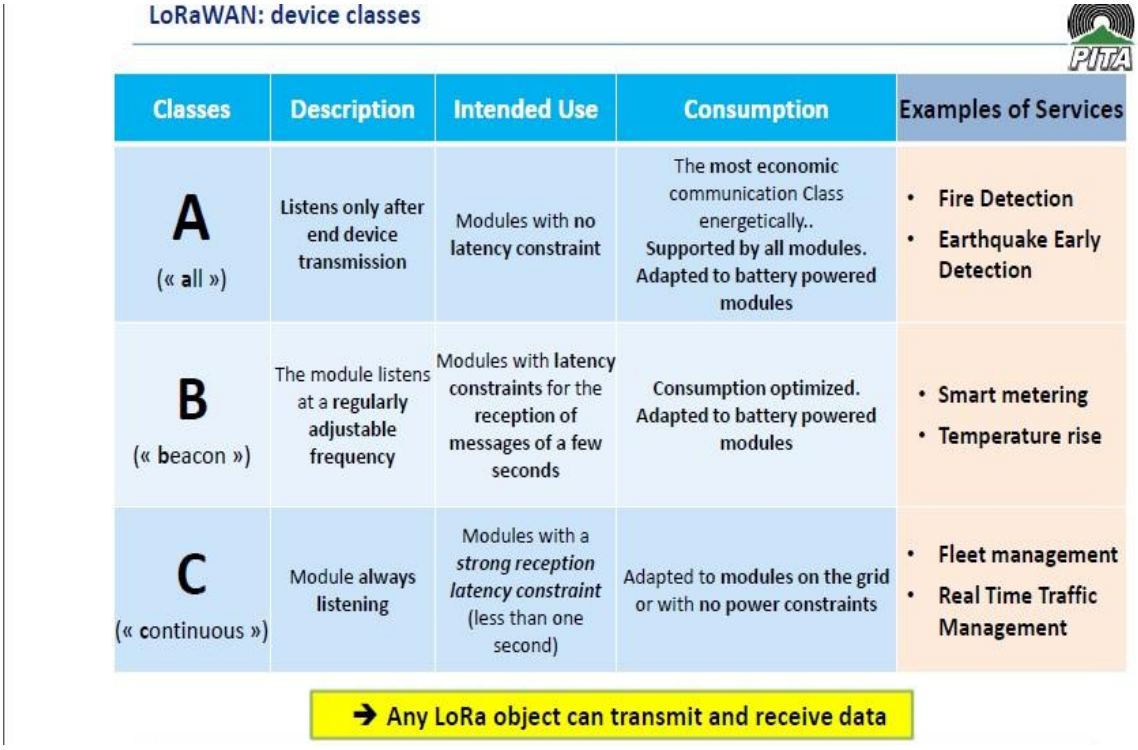


LoRa roadmap

- LoRaWAN is a Low Power Wide Area Network
- LoRa modulation: a version of Chirp Spread Spectrum (CSS) with a typical channel bandwidth of 125KHz
- High Sensitivity (End Nodes: Up to -137 dBm, Gateways: up to -142 dBm)
- Long range communication (up to 15 Km)
- Strong indoor penetration: With High Spreading Factor, Up to 20dB penetration (deep indoor)
- Occupies the entire bandwidth of the channel to broadcast a signal, making it robust to channel noise.
- Resistant to Doppler effect, multi-path and signal weakening.



LoRa Architecture



LoRaWAN: device classes

Classes	Description	Intended Use	Consumption	Examples of Services
A (« all »)	Listens only after end device transmission	Modules with no latency constraint	The most economic communication Class energetically.. Supported by all modules. Adapted to battery powered modules	• Fire Detection • Earthquake Early Detection
B (« beacon »)	The module listens at a regularly adjustable frequency	Modules with latency constraints for the reception of messages of a few seconds	Consumption optimized. Adapted to battery powered modules	• Smart metering • Temperature rise
C (« continuous »)	Module always listening	Modules with a *strong reception latency constraint* (less than one second)	Adapted to modules on the grid or with no power constraints	• Fleet management • Real Time Traffic Management

→ Any LoRa object can transmit and receive data

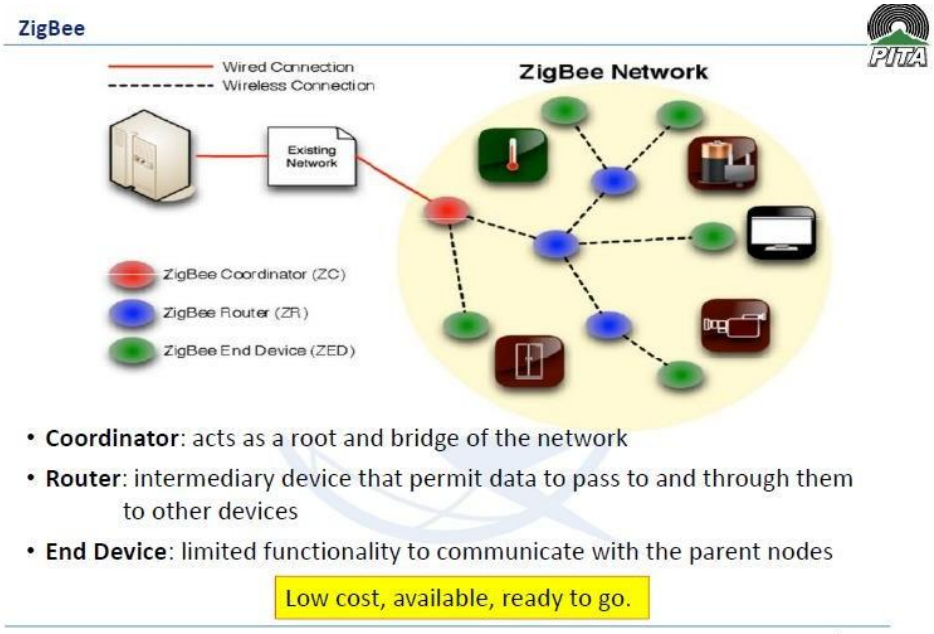
LoRaWAN device classes.

H. Narrow Band (NB-IoT)

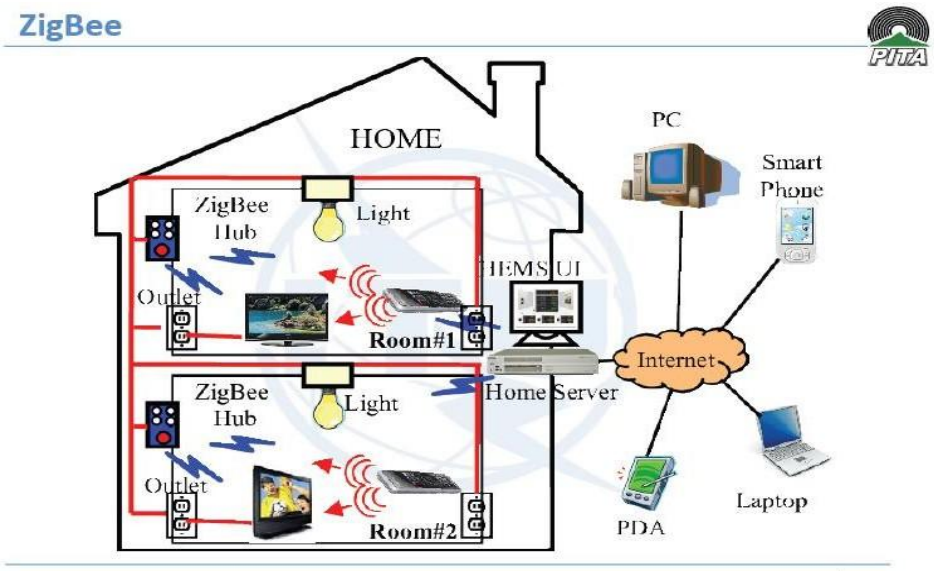
NB-IoT (Narrow-band IoT) is a network that belongs to the LTE-IoT type of wireless network. It is based on the operation of 4G mobile networks. The main features of this network are high coverage, lower power consumption than other LTE-based networks, and even cost, as NB-IoT is more affordable than eMTC. The coverage is large and ranges from 450 MHz to 3.5 GHz. According to Qualcomm, NB-IoT can work on both 2G and 3G networks. The total throughput of the NB-IoT network was reduced to 200 kHz. The NB-IoT network operates on the principle of half-duplex communication and uses TDD (Time Division Duplex) and FDD (Frequency Division Duplex). This divides the available band (into multiple frequencies or time bands) and the transmitter or receiver uses this allocated band to receive or send data.

I. ZigBee

The network originated in the early 21st century in the USA. It was standardized in 2003 by IEEE. ZigBee is a wireless network that has taken its place in the market by its specific features. While other wireless networks have tried for years to introduce new features and improve their performance, ZigBee has remained focused on 8bit microcontrollers. The most famous feature of this network is the long life of the device. The lifespan can be long up to 10 years or even more. ZigBee is similar to Bluetooth technology but has features such as low data rate, and therefore stands high with transfer reliability and good price. Although not used to transmit large amounts of data, some versions of the ZigBee protocol have been modified for such use. Due to its characteristics, it is often used for sensor networks and the transmission distance is up to 100 meters if there is good optical visibility



ZigBee topology



ZigBee home application

J. Cellular

3G/4G/LTE —Carrier broadband has come a long way in the last decade with huge improvements in data handling over mobile operator broadband networks, since the days in the early to late 2000s of 2G and GPRS. GPRS does however have IoT relevance, as it can be useful still in device interconnection in remote areas. However, with 2G networks being decommissioned it might not be around much longer. 3G/4G and LTE have broadband capabilities with high data throughput availability. It is 3G/4G and LTEs wide coverage footprint that makes it so enticing for IoT urban and rural applications. However, it has its drawbacks when used for device connectivity as it requires high power, so has poor battery life. Those failings can be mitigated by connecting many devices to a local 3G router, for example in a hub- spoke topology, which can make better use of the service available. However, because mobile networks are designed for voice and broadband data the price of a data plan and SIM for a remote router can be prohibitive. Mobile operators are beginning to offer IoT service plans so that might change, but even if the price does fall, the inherent design of the networks will not, and that will still make them technically inefficient for devices that communicate only tiny amounts, intermittently.

Edge, Fog Connectivity and Protocols

An edge devise is any piece of hardware that controls data flow at the *boundary* between two networks. Edge devices fulfill a variety of roles, depending on what type of device they are, but they essentially serve as network entry -- or exit -- points. Some common functions of edge devices are the transmission, routing, processing, monitoring, filtering, translation and storage of data passing between networks. Edge devices are used by enterprises and service providers.

Cloud computing and the internet of things (IoT) have elevated the role of edge devices, ushering in the need for more intelligence, computing power and advanced services at the network edge. This concept, where processes are decentralized and occur in a more logical physical location, is referred to as *edge computing*.

One of the most common types of edge devices is an *edge router*. Usually deployed to connect a campus network to the internet or a WAN, edge routers chiefly function as gateways between networks. A similar type of edge device, known as a *routing switch*, can also be used for this purpose, although routing switches typically offer less-comprehensive features than full-fledged routers.

Firewalls can also be classified as edge devices, as they sit on the periphery of one network and filter data moving between internal and external networks.

Championed by Cisco, IBM, and Dell client data is processed at the periphery of the network, as close to the originating source as possible. To enable quicker response, intelligence is pushed from the cloud to the edge, localizing certain kinds of analysis and decision-making. The sheer number of networked devices in the (IoT), growth of mobile computing, and the decreasing cost of computer components all are driving forces behind the move toward Edge Computing architecture. Since transmitting massive amounts of raw data over a network puts tremendous load on network resources, it is much more efficient to process data near its source and send only the data that has value over the network to a remote data center. Time sensitive data will be processed in an edge computing architecture at the point of origin or sent to an intermediary server located in close geographical proximity to the client. Less time sensitive data is sent to the cloud for historical analysis or long-term storage. Edge Computing offers several advantages such as improving time to action, reducing response time down to milliseconds, while also conserving network resources

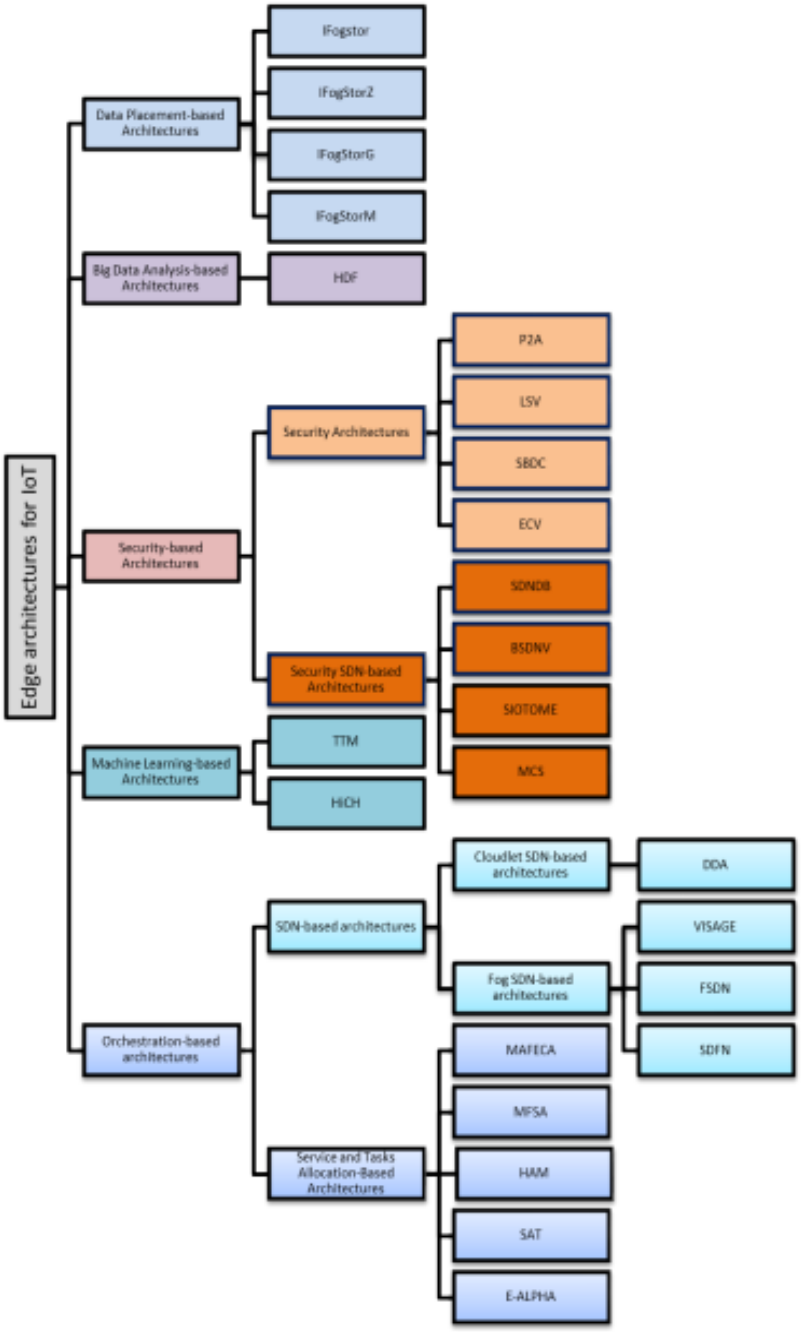


Figure 3. Edge-computing architectures (ECAs)-IoT taxonomy.

Benefits of Edge Computing/Connectivity

Edge for the IoT brings potential benefits for many IoT deployments, including decreased response time along with increased communications efficiency, compared to using the cloud to process and store data. For example, many IoT processes can have a high level of automation at the edge

resulting in low latency for rapid data processing. Only the most important information need then be sent to the cloud for further action or investigation.

Benefits of IoT edge computing that have been identified include:

- *low latency*: By its nature, the edge is closer to the IoT device than the core or cloud. This means a shorter round-trip for communications to reach local processing power, significantly speeding up data communications and processing.

- *longer battery life for IoT devices:* Being able to open communication channels for shorter periods of time due to improved latency, means that battery life of battery powered IoT devices could be extended. distributed ledger, or a hybrid open source ledger implementation such as BigchainDB could be used to obtain the advantage of a distributed ledger which provides features from the NoSQL database MongoDB on which it is based.

- *More efficient data management:* Processing data at the edge makes simple data quality management such as filtering and prioritization more efficient. Completing this data administration at the edge, means cleaner data sets can be presented to cloud based processing for further analytics

- *Access to data analytics and AI:* Edge processing power and data storage could all be combined to enable analytics and AI, which require very fast response times or involve the processing of large 'real-time' data sets that are impractical to send to centralized systems.

- *Resilience:* The edge offers more possible communication paths than a centralized model. This distribution means that resilience of data communications is more assured. If there is a failure at the edge, other resources are available to provide continuous operation.

- *Scalability*: As processing is decentralized with the edge model, less load should ultimately be placed on the network. This means that scaling IoT devices should have less resource impact on the network, especially if application and control planes are located at the edge alongside the data

FOG

Cloud systems are generally located in the Internet, which is a large network of unknown network devices of varying speeds, technologies, and topologies that is under no direct control. As a result, traffic can be routed over the network but with no quality of service measures applied, as QoS has to be defined at every hop of the journey. There is also the issue of security as data is traversing many autonomous system routers along the way, and the risk of confidentiality and integrity being compromised is increased the farther the destination is away from the data source.

IoT data is very latency sensitive and requires mobility support in addition to location awareness. However, IIoT benefits from the cloud model, which handles data storage, compute, and network requirements dynamically in addition to providing cloud based Big Data analysis and real-time data streaming analytics. So how can we get the two requirements to coexist?

The answer is to use the fog.

The fog is a term first coined by Cisco to describe a cloud infrastructure that is located close to the network edge. The fog in effect extends the cloud through to the edge devices, and similar to the cloud it delivers services such as compute, storage, network, and application delivery. The fog differs from the cloud by being situated close to the edge of the proximity network border, typically connecting to a service provider's edge router. It will be on the service provider's edge router that the fog network will connect to, thereby reducing latency and improving QoS.

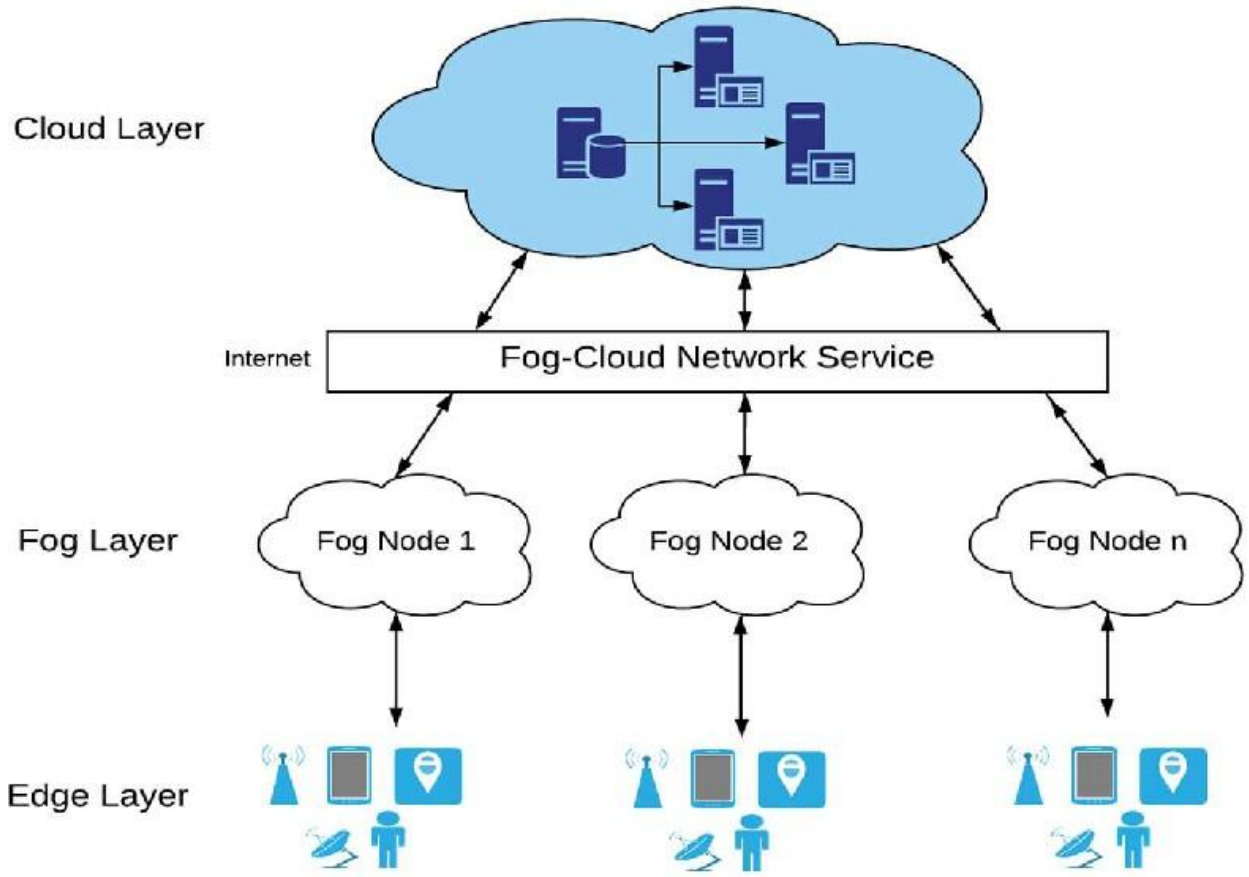
Fog deployments have several advantages over cloud deployments, such as low latency, very low jitter, client and server only one hop away, definable QoS and security, and supporting mobility location awareness and wireless access. In addition, the fog does not work in a centralized cloud location, but is distributed around the network edge, reducing latency and bandwidth requirements as data is not aggregated over a single cloud channel but distributed to many edge nodes. Similarly, the fog avoids slow response times and delays by distributing workloads across several edge node servers rather than a few centralized cloud servers.

Some examples of fog computing in an IoT context are:

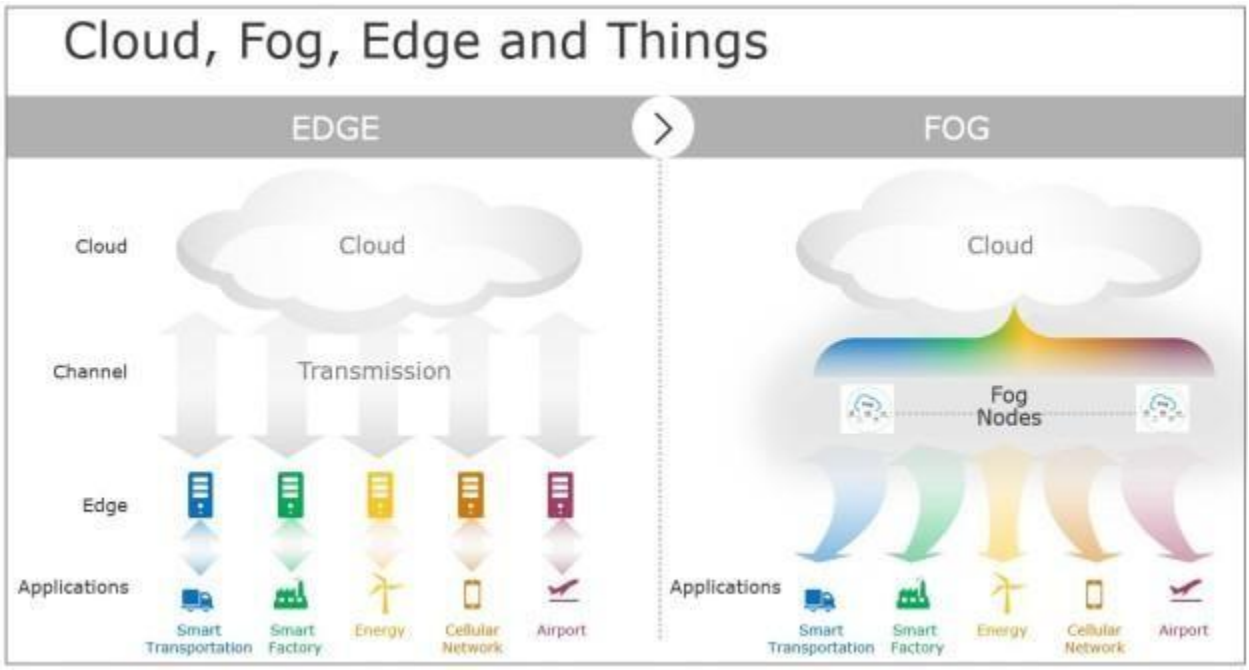
- The fog network is ideally suited to the IoT connected vehicles use-case, as connected cars have a variety of wireless connection methods such as car-2-car, car-2-access point, which can use Wi-Fi, 3g/4G communications but require low latency response. Along with SDN, network concepts fog can address outstanding issues with vehicular networks such as long latency, irregular connections, and high packet loss by supplementing vehicle-vehicle

communications with vehicle-infrastructure communication and ultimately unified control.

- Fog computing addresses many of the severe problems cloud computing has with network latency and congestion over the Internet; however, it cannot completely replace cloud computing which will always have a place due to its ability to store Big Data and perform analytics on massive quantities of data. As Big Data analytics is a major part of the IoT and then the cloud, computing will also remain highly relevant to the overall architecture.



Three Layer Edge-Fog-Cloud architecture



Two Layer Cloud-Edge and Cloud-Fog architectures

Protocols

There are many industry protocols that facilitate different styles of device communication. An edge solution should support the most common protocols. For example, these include Z-Wave, ZigBee, Bluetooth LE, etc.

DDS/RTPS as Edge/Fog computing protocols

The main Edge/Fog computing goal is *minimizing latency and save bandwidth*, adding some intelligence to the access point to optimize the typical scenarios in the IoT. Examples of Edge/fog computing scenarios include low power devices, real-time requirements, and wireless networks.

Taking into account these requirements, *a good Edge/Fog computing protocol should be lightweight, transport agnostic and customizable*. Heavyweight Protocols such as Web Services or REST are discouraged because consume a lot of resources in terms of bandwidth and CPU use. Remember the IoT devices include small sensors, embedded devices and low bandwidth data links.

The Data Distribution Service (DDS) middleware standard is widely known for being used in very exigent real-time distributed systems in Aerospace and defense systems, and its underlying protocol RTPS (Real Time Publish Subscribe Protocol) main features make it very suitable for the goals of Edge/fog computing:

- Designed for **real-time** distributed systems: low latency communications and low bandwidth use.
- **Highly customizable**: It allows to setup different Quality of Service parameters for different kinds of data or different data links.
- **Wireless enabled**: It can be deployed over any transport/data link, including specifically disconnected and intermittent data links, such as low bandwidth wireless networks. RTPS supports multicast, a very recommendable feature in this kind of networks, and it can also be deployed over non IP-Based data links.
- **Lightweight**: It requires very few resources, and it can be implemented in low power devices.

Signal Processing

The signals carrying IoT data are highly likely to face numerous obstacles and can be corrupted by significant amount of noise present in the environment. White Gaussian model has commonly been used to quantify the noise. The types of noise which have been found to degrade the quality of IoT signals vary from the impact noise resulting from high frequency interference and instantaneous disturbance on the initialization of large equipment to changing connections around the participating IoT devices. All these kinds of noise negatively influence the multi-device information fusion system.

Such noises should be filtered out and the transmitted signal should be reconstructed back to its actual form to ensure the accuracy and reliability of the transmitted information. Here, accuracy of IoT solutions is measured in terms of the number of packets reporting correct information, deviation between the reported and actual results and the delivery to correct destination timeliness.

Assuming that the received signal is affected by white Additive Gaussian noise (AWGN) which is also *stationary* in nature, the received signal $y(i)$ can be represented as,

$$y(i) = x(i) + \sigma\epsilon\,(i),\ i = 1,2,3\ \ldots n-1$$

where $x(i)$ is the noise-free transmitted signal, $\epsilon(i)$ representing independent normally distributed random variable and σ representing the intensity of the noise affecting $y(i)$.

Reconstruction of the original signal $x(i)$ from the instantaneous set of $y(i)$ values without actual assuming a specific model for $x(i)$ or $y(i)$ is the primary aim of the process of '*Denoising*'. The most common approach is to recognize noise components as the high frequency components present in the corrupted received signal, apply Fourier transform and then filter out the high frequency components.

Therefore, the most traditional way of denoising signals is based on Fourier analysis and Fourier transform.

Wavelet Transform (WT)

Wavelet Transform (WT) is another method used in denoising signals. On contrary to the traditional Fourier transform, WT is particularly suitable for application of *non-stationary* signals which may instantaneously vary in time. Primarily, the received signal is divided into different frequency components using wavelets. The basis function of WT is scaled based on frequency and a subset of small waves (known as mother wavelet) is used for implementing

WT. The mother wavelet is a time-varying window function used for decomposition of *x(i)* into weighted sets of

scaled versions of *y(i)*. Consequently, using wavelet transform in signal processing is the process of the partial transformation of the spatial domain and the frequency domain, in order to get useful information accurately from it though corrupted with noise.

Since different frequency levels are used for WT, it is quite convenient for analyzing the signal characteristics at different frequencies and detecting removing corrupting noise. Broadly, there are two types of WT, Continuous WT (CWT) and Discrete WT (DWT).

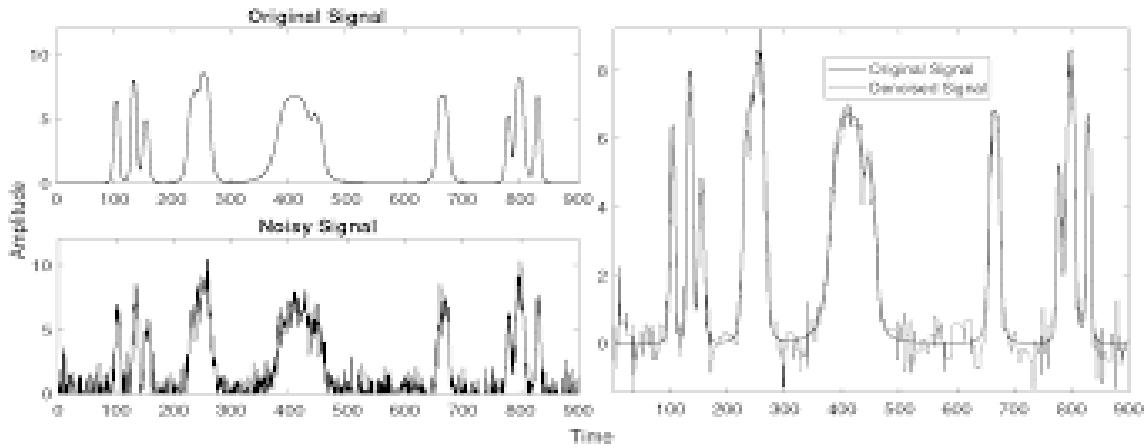
Discrete Wavelet Transform for denoising data

The DWT denoising procedure consists of three steps. In the first step, if the length of the data stream is of length of the order of power of two, it is transformed to the wavelet domain. In the second step, coefficients with either zero magnitude or criterion-based minimized values are selected. In the third or final step, the minimized coefficients are reverted back to the original domain from the wavelet domain to extract the denoised data. DWT-based denoising techniques can be broadly classified into two categories - linear and non-linear.

In linear DWT, signal and noise are assumed to be belonging to the smooth and the detailed part of the wavelet domain, where high frequency components are attenuated. While in nonlinear DWT, the filter removes the coefficients selected in the second stage with amplitudes less than the threshold. In practicality, non-linear DWT is always preferred over linear DWT, as linear DWT introduces error due to the retention of noise components and loss of signal components owing to wavelet filtering.

Whether linear or non-linear DWT denoising technique is used, performance depends on the choice of the wavelet family and the length of the filter. The traditional way for making this choice is based on visual inspection of the data, for example, daublets are implemented when the data appears smooth in the wavelet domain, while Haar or other wavelets are used when the data appears bursty and discontinuous in the wavelet domain.

In order to overcome the problems with DWT denoising, correlation denoising method was introduced. Correlation denoising method implements wavelet transformation and filtering in a way such that the correlation between wavelet coefficients of the signal part and the noise-part is different at each level. However, correlation denoising in its original form is computationally complex. In order to reduce computational complexity, wavelet threshold denoising method was proposed by D. L. Donoho. The method is simple to calculate and the noise can be suppressed to a large extent. At the same time, singular information of the original signal can be preferred well, so it is a simple and effective method. A brief overview of what happens when DWT is applied for denoising is demonstrated in the diagram below.



The four major components of the DWT denoising technique are: wavelet-type selection, threshold selection, threshold function selection and threshold application to the wavelet coefficients.

1. **Wavelet Selection** - There is a wide variety of wavelets that can be used for denoising. Selecting the optimum one depends on the selection of the matching wavelet filter. Out of different wavelet transform based denoising methods, only minimum description length (MDL) method has the flexibility of choosing the filter type.

2. **Threshold Selection** - There are four basic types of threshold selection, minimax, Stein's unbiased estimate of risk (SURE), and minimum description length (MDL). The ***Universal*** threshold is computed using,

$$t = \sigma \text{ X } \sqrt{(2 \text{ X } \ln(N))}$$

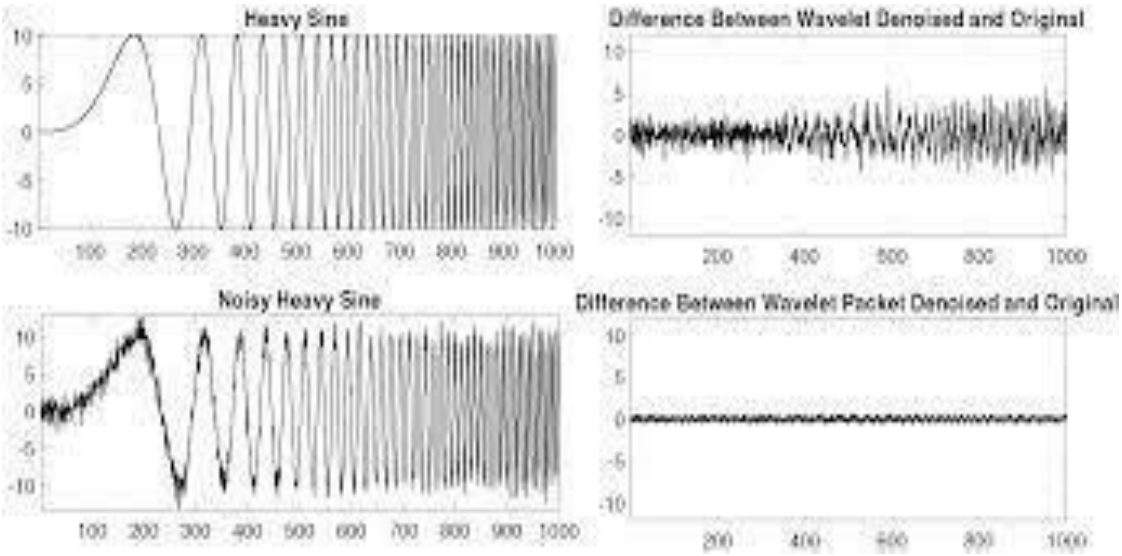
for which N is the length of the signal data array, and σ is the standard deviation of noise. In practicality, in most cases, σ is unknown, but can be estimated using the first detailed part of the wavelet coefficient x_i through the expression,

$$\sigma_{estimate} \approx \frac{\text{Meridian}(|x|)}{0.6745}$$

Signal denoising for IoT networks

The huge amount of sensor data generated in an IoT network are used to take decisions on a certain observation/ phenomenon based on real-time processing. The decision-making procedure often involves detecting the signal energy level transmitted from the sensors. If the received energy level is higher than a predefined threshold, the target phenomenon is detected to be present and vice- versa. However, the sensor data gets crippled with noise contributed by the wireless environment

and the internal electronics of the sensors, on its way to the data center for processing. The WPT method will be the best option in this case for denoising the sensor data, where the original signal coefficients are preserved while removing the noise within the signal. The WPT method can decompose a signal in both scale and wavelet space thereby revealing more details about both the sensor signals and the crippling noise. If energy correlation analysis is used in conjunction with WPT, signal energy from the sensor data can be analyzed and noise can be eliminated by zooming into the signal characteristics at different time scales. Advantages of WPT over WT is evident in the figure below**.** Hence, in this section, a universal framework is presented for denoising sensor signals in IoT networks. The framework is based on energy correlation analysis and combines the processes of WP decomposition, coefficient modification and WP reconstruction.



A key aspect of IoT is event-driven or a periodic sampling. Traditional digital signal processing and control assume periodic samples resulting in time-series data. However, time series consume too much power at the nodes and too much bandwidth on the network. Not all applications are amenable to aperiodic data acquisition.

Constraints on power and bandwidth also encourage distributed computing over sensor events. Relatively small processors can perform useful processing on many data streams. Recognizing interesting events using edge processing reduces the amount of network bandwidth consumed; it also reduces power consumption since wireless communication requires large amounts of power. Cloud computing (centralized servers) or fog computing (servers closer to the edge) can be used to perform further processing on those extracted events.

Data Analytics

Real-Time and Streaming Analytics

When the data size is massive, generally it is being subjected to batch and historical processing techniques to extract all kinds of hidden intelligence. However, on the other hand, there are fast and streaming data which are relatively small in size but widely vary in structure and speed. There are open source and commercial-grade platforms emerging and evolving for undertaking real-time and streaming analytics on fast and streaming data. Recently, there has risen demand for doing immediate and interactive analytics not only on fast data but also on big data to emit out insights in time. With the grand availability of exemplary solutions, systems, and services, the days of accomplishing real-time analytics on big data are not too far away. All these advancements do not imply that the fully matured and stabilized BI and DW solutions are going away. But there is a telling need to complement them with newer technologies for big, fast, streaming, and IoT data storage, management, and analytics. The era of real-time and streaming analytics is steadily arriving, and there are a number of use cases being unearthed mandating real-time and streaming analytics capabilities and competencies for any organization to grow fast. There are well-known use cases such as full-text indexing, recommendation systems (e.g., Netflix movie recommendations), log analysis, computing web indexes, and data mining. These are the well-known processes that can be allowed to run for extended periods of time

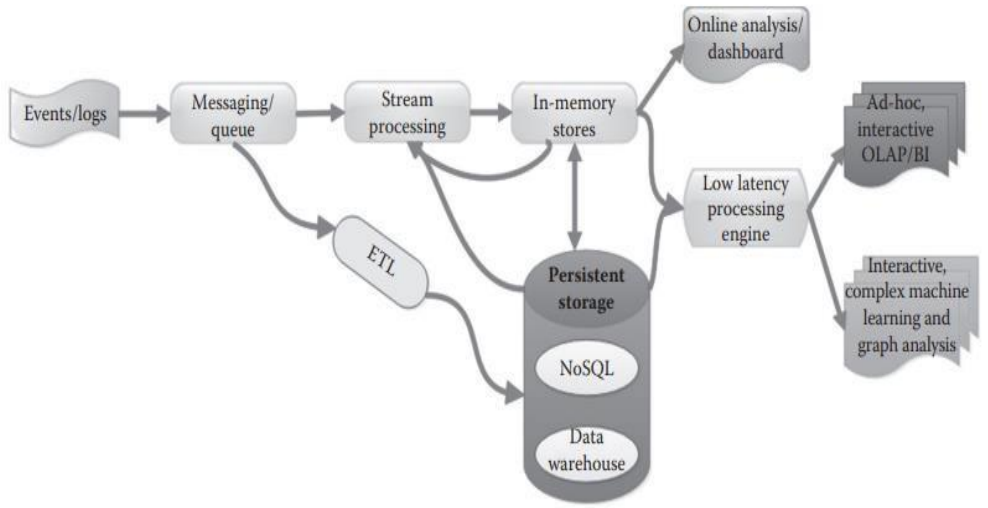


Figure 7.2 The real-time and streaming analytics architecture.

Operational analytics (OA) is a kind of real-time analytics that provides enhanced visibility into business processes, events, and operations as they are happening. The practice of OA is succulently enabled by special technologies that can handle machine data, sensor data, event streams, and other forms of streaming data and big data. OA solutions can also correlate and analyze data collected

from multiple sources in various latencies (from batch to real time) to realize and reveal actionable information. Organizations can act on extracted knowledge by sending an appropriate alert notification to the correct manager in time, updating a management dashboard, offering an incentive to a churning customer, adjusting machinery, or preventing fraud.

Data Analytics Platforms

These are a bit complicated platforms directly attached with AEPs. There are powerful data mining, fusion, and processing algorithms getting implemented and incorporated in these platforms to elegantly draw knowledge out of data heaps. The insights extracted are then fed into AEPs to drive pioneering applications. The challenge with these platforms is that the amount of sensor and actuator data is becoming massive, and hence doing real-time analytics on big data is a bit difficult. There are technologically sound and solid solutions emerging in order to tackle this particular problem. The underlying IT infrastructures also need to be exceptionally elastic enough to cope with the big data challenges. Increasingly data storage, processing, mining, and analyzing software solutions are being deployed in private, public, or even in hybrid clouds. Data are subjected to a variety of deeper and decisive investigations in order to extract pragmatic insights. There are big, fast, streaming, and IoT data whereas there are batch or ad hoc or historical or comprehensive processing apart from interactive and iterative processing. Thus, there are specific as well as integrated platforms on cloud environments for embarking on big and fast data analytics. Especially predictive, prescriptive, and personalized analytics domains are expected to be dominant in the years ahead.

Local Analytics

Local analytics, entails processing some analytical information at the gateway, so that only meaningful information is sent to the next tier which could be another gateway, the data center, or the cloud. This minimizes consumption of expensive network bandwidth and reduces overall solution latency.

Cloud Analytics

Cloud analytics is the use of remote public or private computing resources—known as the cloud—to analyze data on demand. Cloud computing analytics helps streamline the business intelligence process of gathering, integrating, analyzing, and presenting insights to enhance business decision making.

How cloud analytics works: Cloud analytics works by allowing a business to use the advanced data analytics tools available on cloud analytics platforms to analyze vast quantities of data. Businesses can then report and store those findings for repeat use. Cloud analytics offerings are typically offered as a subscription or pay on a volume of data or query basis. Cloud analytics has proven to be a faster way to gain business-critical insight for decision making.

Advantages of cloud analytics: Advantages of cloud analytics include helping businesses more efficiently process and report data findings, enhance collaboration, and provide decision-makers faster access to business intelligence.

Scalability & agility: Scalability, in volume of users and data, is one of the advantages of cloud analytics. Another is the increased agility to ask new questions on demand. Cloud computing can enable the abilities to add data storage and data analysis capacity as needed. In this way, businesses can most efficiently scale the storing, processing, and leveraging of data through insights that reflect changing market conditions.

Unified approach to data: Cloud analytics can provide a unified approach to data, allowing fast access to necessary business data by your organization's decision-makers. That includes access to different types of data, as well as data from different sources. You can rapidly explore, test, and evaluate data to find strategic insights that advance the enterprise.

Breaks down silos: When a business takes advantage of cloud analytics, it can effectively achieve cross-organizational integration of data. Leveraging data from different parts of your organization through a cloud analytics solution can deliver daily insights. These insights can be a competitive advantage that propels your business forward. Data stored and analyzed in the cloud also makes it easier for employees of all types to quickly access data and share insights, supporting effective communication, collaboration, and decision making.

Find answers and draw insights quicker: A cloud analytics platform allows businesses to better integrate—and analyze—data quickly to identify intelligent insights for actionable decisions at the moment of relevance.

Encourage more collaboration: When businesses use a cloud analytics platform, they gain a central place to access data and connect to shared information whenever necessary or proactively delivered. This shared connection and fast access to data enhance collaboration throughout the enterprise.

Increase data security and governance: A cloud-based analytics platform can provide a single and secure access point to data. Also, a cloud analytics platform improves data governance by giving you more granular access controls for who has access to what data and audit capabilities to understand who accessed what data.

Legal Challenges in IOT

Vulnerability Concerns and Groundwork for Legal Challenges

Despite the many benefits of IoT, increased connectivity between devices and the Internet tend to create an array of security and privacy risks. The complexity and connectivity commonly found in IoT introduces new security vulnerabilities. Government and corporates are worried that the proliferation of cheap IoT devices and their applications can create security issues for consumers in terms of data collection and purpose of use.

IoT devices are prone to many vulnerabilities as they can be used to direct attacks (such as denial of service) on the consumer's network. Smart televisions store or transmit information that could be exploited by hackers to compromise personal information. A hacker could attack one of the consumer's interconnected devices which not only provides him with access and control of the devices such as video cameras, sensors, or smart light bulbs, but also provides him with the opportunity to gather all manners of personal information. The reality is that consumers' concerns regarding their safety, privacy, and data security trump their fascination with IoT devices.

The U.S. Department of Homeland Security (DHS) asserts there is no clear understanding of who is responsible for security decisions in a situation in which one vendor designs a device, another supplies component software, another organization operates the network in which the device is embedded, and another deploys the device. It argues that this challenge is amplified by lack of comprehensive widely adopted international norms and standards for IoT security, lack of incentives for developers to adequately secure products and unequal awareness of the methodology of evaluating security features of competing options. Digital goods have moved away from the traditional legal models as IoT manufacturers and distributors are gradually shifting the rules of ownership and as such, there is a blurry line as to who actually owns our IoT devices such as phones, refrigerators, cars and televisions. The question asked is whether the purchasers are merely licensing these devices. This issue has generated different schools of thought. While some IoT device manufacturers assert that it is illegal for anyone other than the device manufacturer to diagnose and fix the problem when our devices breaks, others claim that they own any data generated by the device and not the users who purchased the device and created the data. As user and consumer advocates continue to polarize these assertions, device manufacturers continue to advance this cause.

The battle of who "owns" the device continues to gather momentum with Apple insisting that "iPhone users are licensees, not owners, of the copies of the iPhone operating software". Elsewhere, vehicle manufacturers such as John Deere, Ford, General Motors, Mercedes Benz, and

Ferrari have resorted to using technology and law to weaken property interests of drivers. For example, newer John Deere tractors contain at least eight control units – software and hardware components that regulate various functions that no longer allow farmers to maintain and repair their own equipment other than to send them to the factory. The tractor manufacturer is said to have interposed the firmware between the farmers and the tractor, claiming ownership of the vehicle while relegating the farmers as merely having "an implied license for the life of the vehicle to operate the vehicle". In 2015, Ford recalled 433,000 new cars and SUV vehicles due to software glitches that could allow the vehicle to continue running the engine after turning the ignition key to the "off" position, and which could also allow the vehicle be rolled away or be stolen. In 2015, Chrysler recalled 1.4 million vehicles due to hackable vulnerability on the vehicles' Uconnect dashboard computers which could result in the hackers taking over the dashboard functions, steering, brakes and transmissions.

The vulnerability of IoT devices and products were actually encouraging attacks against consumers. A 2015 Hewlett Packard report on home security systems (motion detectors and video cameras) showed that 100 percent of the devices studied contained significant vulnerabilities such as password security, encryption, and authentication issues. Its report the previous year had indicated 70 percent of IoT devices were susceptible to attacks such as password security, lack of user access and encryption.

Ukraine's power grid has been attacked twice in between 2015 and 2016. The 2015 cyberattack on Ukraine's power grid was as a result of BlackEnergy malware found on the computer systems of the affected power companies and caused power outage for several hours. The hackers had used malicious firmware to replace the legitimate firmware in the serial-to -Ethernet converters at many substations which gave them access to SCADA network through the hijacked Virtual Private Network. This allowed the hackers to successfully launch commands that disabled the Uninterruptible Power Systems they had earlier reconfigured while rendering the converters inoperable and unrecoverable. The DHS attributed the malware as being directed to attack Ukraine's obsolete and vulnerable industrial control systems.

The damage could have been worse and Ukraine's saving grace was the antiquated technology was not Internet connected, allowing companies to restore power within a short period of time. The second Ukraine attack in 2016, which also resulted in power disruption was allegedly masterminded by Russian hackers who compromised some computers at an electricity control center to plunge parts of the city of Kiev into darkness. The hackers were said to have sent malware to the employees via e-mail which in turn facilitated the stealing of the login credentials to shut down the substations.

Liabilities and Class Action Suits

In March 2015, the first IoT class action was instituted against three giant automakers – Ford Motor Company, General Motors and Toyota Motor Corporation at the U.S. District Court for the

Northern District of California on the backdrop of a report released by U.S. Senator Edward Markey.

In Cahen et al. vs. Toyota Motor Corporation, 147 F.Supp.3d 955, N.D. Cal. It was alleged that some models of their vehicles had defective electronic systems that were vulnerable to remote hacking and which had a potential of compromising safety and personal information. Investigations had revealed that hackers could hijack the controls of the affected vehicles from the driver to cause unexpected acceleration, loss of breaks, turns, horn activation, modification of the speedometer and gas readings. However, the case was dismissed and also upheld at the 9th Circuit of the U.S. Court of Appeals based on jurisdictional grounds and lack of standing as the plaintiffs could not prove any actual injury or any immediate threat of injury.

In July 2015, Fiat Chrysler faced a class action suit in the U.S. District Court for the Southern District of Illinois after recalling 1.4 million of their vehicles affected by a hackable software in Chrysler's UConnect dashboard. The UConnect consists of software and hardware components which control Bluetooth phone, navigation and entertainment systems. It was alleged that hackers could gain control of the radio and air-conditioning features and the entire vehicle via the vehicle's input and USB port. The publication of vulnerabilities of automobiles to Internet hacking gingered U.S. Senators Edward Markey of Massachusetts and Richard Blumenthal of Connecticut to introduce "legislation that would empower the US National Highway Traffic Safety Administration (NHTSA) and the Federal Trade Commission to secure vehicles from hacking threats and main driver privacy".

Other IoT-based class action suits have followed. In 2015, a class action suit was brought against Mattel, ToyTalk, and kidSAFE – makers of the high-tech, talking Barbie doll by parents who claimed that a security flaw with the doll allowed hackers to compromise their children's privacy. The parents alleged that the 'Hello Barbie' doll collects personal data that could be easily compromised by hackers. VTech, another toy manufacturer was hit with a class action suit in 2015 over its failed effort to safeguard personally identifiable information (PII) of about 10 million American children and the parents. The hackers compromised personal information (names, birth dates, gender, e-mail addresses and profile names) of the children and their parents. VTech provides the software that allows parents to limit children's Internet access and to track their learning achievements through the "Learning Lodge"

Deconstructing the Legal Challenges

To illustrate how IoT controls our daily lives, an examination of a Breathometer, which was a device used to measure the blood alcohol content of users by using their iOS (iPhone operating system) and Android smartphones is helpful. Breathometer is representative of devices that promise efficiency, social, and individual benefit via quantification and monitoring of once immeasurable qualities. A consumer purchases this device, unpacks it, and reads the user's manual that elaborates how to download the Breathometer app, creates an account with the company via

the app, and then plugs the device into her smartphone. However, the manual does not contain any information about privacy policy that might apply to the data generated by the device. Nor is there any description or explanation of what data the device will generate, where such data will be stored (such as user's phone or company's server in the cloud); whether such data can be erased and how; how the company might use such data, if the company will sell it or could it be required by a court? When installing the Breathometer app through the Apple App Store, there is no mention of any privacy policy. No pop-up message with such a policy appears when the user creates her account via the app or starts using the device.

It is only when the purchaser then logs into the company's website, scrolls down the bottom and clicks the small link for "Privacy Policy" can she learn that her blood alcohol test is being stored indefinitely in the cloud, and cannot be erased by the user, may be disclosed in a court proceeding if necessary, and may even be used to moderate advertisements at the company's discretion. Has anyone pondered the employment decisions; criminal liability implications; and health, life or insurance ramifications this data could generate? How much power and information has the user or consumer given up to the Breathometer company?

On the surface, the consumer has been misled to assume the Breathometer is a small black device like any other device, not knowing it is a data source and cloud base repository. Now, imagine when the device is stolen or the data generated and stored on this device falls into the hand of criminals. The description of the Breathometer highlights the real challenges of IoT in terms of what information do these devices collect, how would such information be used, what real choices do consumers have about such data, and who owns the data?

Current Legal Problems

It is evident that consumer devices such as home and appliance monitors, automobile black boxes, personal health monitors, smartphones and employee monitors generate huge amount of valuable data about their users' activities, habits, personalities, preferences and attributes. This goes to expose the grave dangers of these IoT devices despite their flurry of benefits to mankind.

The use of IoT devices presents four legal problems - (1) discrimination, (2) privacy, (3) security, and (4) Consent.

1. *Discrimination*

The widespread deployment of sensor poses both technical and legal problem for IoT because IoT data allows consumers to be sorted and discriminated against. Data collected by IoT devices can be used for both beneficial and discriminatory purposes as exemplified in the case of personal wearable fitness tracker which continuously tracks detailed information about the person's movement and biometric data. This data can be analyzed to determine the person's level of fitness, estimate calories burned, track hours left, which then will be used to characterize the quality of sleep and sleep pattern. The data can also reveal the person's disordered eating behavior, insomnia,

diabetes, cigarette smoking, or alcohol and drug abuse.

The sensor data from IoT devices can provide unexpected information about individual consumers which allows employers, lenders and insurers use this data to draw inferences and make economically important decisions, without the industry regulators or consumers having knowledge or understanding of the process. The data can provide huge amount of the person's information to her prospective employer or insurance company which may also be used to deny her employment or application. The current antidiscrimination and credit reporting law is unprepared to handle IoT-based discrimination in which consumer information is revealed. Two legal concerns arise from the possibility of indiscriminate exposure of information.

- First, will IoT lead to new forms of discrimination against protected class, such as race, age or gender?
- Two, will IoT lead to disturbing forms of economic discrimination? This type of discrimination will open doors to new patterns of illegal discrimination against age, race or gender and may also trigger class action lawsuits.

2. Privacy

Privacy is meant to respect customer's information and their rights to determine how such data is used. Accountability, a key privacy principle is difficult to keep with IoT generated data because of the involvement of multiple stakeholders such as device manufacturers, third party applications and social platforms who collect, use or exposed data. IoT can jeopardize a person's expectations of privacy in many ways as shown in Riley v. California.

The case raised a common question of whether the police may, without a warrant, search digital information on a cell phone seized from an individual who has been arrested. Riley was stopped by a police officer for driving with expired registration tags. In the course of the stop, the officer also learned that Riley's license had been suspended. The officer impounded Riley's car per department policy, conducted a search which revealed possession of concealed and loaded firearms and two handguns. Riley was arrested and his cell phone was seized and upon examination of its content, the police found pictures of Riley standing in front of a car they suspected had been involved in a shooting a few earlier. Riley was finally charged, in connection with earlier shooting, with firing at occupied vehicle, assault with a semiautomatic firearm, and attempted murder.

The U.S. Supreme Court ruled that information on cell phone is not immune from search, but the Police must obtain a court search warrant before searching a cell phone seized incident to an arrest. Although Justice Alito concurred to the decision, he stated that the rule of pre-digital era should not be used in searching a cell phone and called for a new balancing of law enforcement and privacy interest. His opinion was that privacy protection in the 21st century should not be left primarily in the hands of the federal courts using the blunt instrument of the 4th Amendment, citing it as the prerogatives of Congress and state legislations

It is clear that Riley's cell phone information linked him to shooting and attempted murder charges and also led his 4th amendment violation. Mobile phones contain huge amount of data such as videos, photographs, phone book, calendar, text messages and Internet browsing history, bank statements, medical information, and other sensitive documents which have interrelated consequences for privacy. Although IoT devices such mobile phones, digital cameras and GPS can be beneficial to law enforcement agents in tracking criminals, the manner they are used can also violate consumer's privacy (Rose et al., 2015). The problem is how to encourage or compel IoT manufacturers to integrate privacy-by-design and security-by design principles into their core values and how device functionality and privacy requirements can be reconciled; Congress is needed to enact appropriate legislations in this direction.

3. Security

Data collected from IoT devices may not be restricted from being sent across jurisdictional boundaries as these devices communicate via the Internet, and the Internet spans different jurisdictional boundaries. IoT devices can collect people's data in one jurisdiction, transmit and store that information in another jurisdiction for processing and storage. This can become a legal problem if the data collected is private and sensitive and subject to data protection laws in multiple jurisdiction. Encryption may be a requirement in one jurisdiction but not required in another jurisdiction.

For example, consider a Fitbit monitoring device collecting a consumer's information in the United States and storing the data in the organization's Web server in Switzerland. More complications arise when the data protection laws in United States where the device and data physically reside may not be consistent or compatible with the laws of Switzerland where the data is stored and hosted on the Web server. Trans-border data flows raise questions about the legal scope of regulations that may be applicable. In this situation, it may be difficult to isolate which law governs the device collecting the data in USA, and which law that governs the storage and use of the collected data in Switzerland. The question becomes whether a jurisdiction with more-restricted data protection (such as encryption and strict access controls) laws for handling and transmission of IoT-enabled data be able to project such legal requirements onto other jurisdictions. This is a new challenge that IoT devices pose because the devices can automatically connect to other devices and systems and transmit data across international borders without the knowledge of the user. This creates a liability for cross-border data flow requirements.

IoT devices may be prone to security flaws and the devices may not be patchable or easy to update and thus get exposed to hacking; insulin pumps have been reported to be exposed to hacking. Data security law is unprepared for IoT security problems. For example, in US, the FTC enforcement and state data breach notification laws regulate data security in the United States. Neither of these laws is clearly applicable to breaches of IoT data as there is no general federal data-security statute yet, prompting FTC to use its general authority under the FCT Act to institute penalties against companies with security negligence. There are also considerations of IoT security

violations under state data-breach notification statutes.

4. Consent

Discrimination, security and privacy issues about IoT underline the new and exclusive ways in which connected sensor devices could harm the wellbeing of consumers. IoT devices complicate consent as they do privacy, security and discrimination. Consumer protection laws associated with privacy-policy disclosures is currently not prepared to tackle with these issues. Sensor devices pose a threat to consumer consent. IoT devices are screen-less and lack an input mechanism such as a keyboard or touch screen. IoT device manufacturers prefer to provide privacy and data-related information in website privacy policies. The ambiguity of IoT privacy policy language further compound issues in the sense that there is no clear policy understanding of whether sensor or biometric data count as personal information and how such data can be shared with or sold to third parties. Consumer protection law is unprepared as most regulator's authority is to guard deceptive and unfair practices. IoT data are difficult to anonymize and secure and this creates some privacy issues

IoT Device Liability and Concerns Regarding Data Security and Privacy

IoT devices pose legal liability concerns that need consideration. The fundamental question regarding IoT devices is: Who is responsible if a person is harmed as result of an IoT device's action or inaction? There is not much case law to support this position which has led to a barrage of questions. Currently, there is no defined statute as to who owns the data generated and collected by IoT devices – consumers or device manufactures? There are also concerns as to who is responsible or liable for patching IoT devices, routers, switches and cloud connections as well concerns if the user can view, edit, or delete sensor data from the manufacturer's servers if it is stored there. Does the IoT device have security controls built-in? Is device information encrypted and how?

Questions are asked about what exact information the device will collect about itself or its user, using what sorts of sensors? Is that information stored in the device itself, on the user's smartphone, on the manufacturer's servers in the cloud, or all of the above? Where will the information be stored? Does the device support an industry or regulatory function? What data will the device disseminate and what kind of security controls can the IoT device support?

Consumers wonder if there are situations where IoT devices should not be collecting data and who exactly will the manufacturer or service share the data with, and whether the user have any right to opt out of such disclosures? Can the user gain access to the raw sensor data in order to export it to another service or device and what happens if the IoT product vendor goes out of business or no longer supports the product? What happens when the Internet connection goes down?

Issues have also been raised regarding if a cloud-based IoT device termed "remote computing service" is protected by the data protection Act? What is the classification of the data and risks

associated with it being transported? What is the intended useful life of the device? Are proprietary accesses needed for updates and maintenance? These are legitimate concerns and questions that continue to trail the adoption, growth and development of IoT unless legislative bodies and device manufacturers properly address them.